

Evidencia Situación problema Series Tiempo - Capital Analyst

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Series de Tiempo

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Evidencia Series de Tiempo

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Estrategia de Portafolio: Página 45-46.

Evaluación de Resultados y Reflexiones Finales: Página 46.

Interpretación de las Pruebas de Hipótesis, Conclusiones y Recomendaciones: Página 47.

Link a colab:

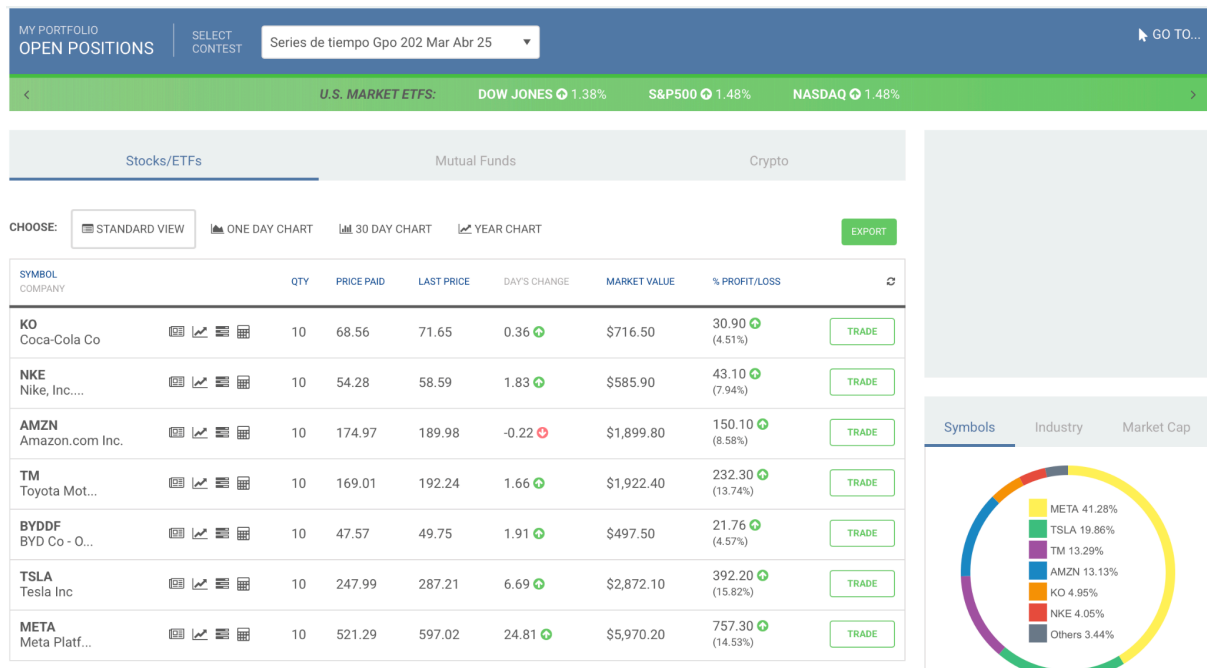
<https://colab.research.google.com/drive/1MJlvKeO4pEiAzJdLVbHmtp7mWm32yZQe#scrollTo=hgBdfJGP184&line=251&uniqifier=1>

Link a Grok:

https://grok.com/share/c2hhcmQtMg%3D%3D_bc215cb3-18a5-4e52-a3b4-110f39fdf269

Link a Github:

<https://github.com/a01741844/Evidencia-Series-de-Tiempo.git>



Interpretación:**AMZN:**

Amazon mostró no estacionariedad tanto en 5 como en 30 minutos, lo que la hace poco predecible. Sus modelos ARIMA no son tan estables y la varianza es alta. Por ahora prefiero no invertir mucho o buscar oportunidades en corto.

KO:

Coca-Cola resultó ser la más estable, especialmente en 5 minutos. La estacionariedad es clara y los modelos ajustaron bien, con varianza baja. Por eso, creo que es una buena opción para comprar en el corto plazo y aprovechar movimientos sostenidos.

BYDDF:

BYDDF se ve bastante volátil. Aunque tuvo cointegración con Amazon, no es suficiente porque la serie no es estacionaria y los modelos muestran heterocedasticidad alta. Me da poca confianza entrar ahora invirtiendo mucho y preferiría esperar o buscar cortos.

NKE:

Nike tiene una tendencia fuerte aunque no es estacionaria. Los modelos ARIMA ajustaron bastante bien y la autocorrelación es alta. Me parece una buena opción para seguir la tendencia si el mercado sigue empujando, aunque cuidando mucho la entrada.

Resultados Programa:

Please upload AMZN 5-min Excel file

AMZN 5 min.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) -

25629 bytes, last modified: n/a - 100% done

Saving AMZN 5 min.xlsx to AMZN 5 min (1).xlsx

Please upload KO 5-min Excel file

KO 5 min.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 25325

bytes, last modified: n/a - 100% done

Saving KO 5 min.xlsx to KO 5 min (1).xlsx

Please upload BYDDF 5-min Excel file

BYDDF 5 min.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) -

15390 bytes, last modified: n/a - 100% done

Saving BYDDF 5 min.xlsx to BYDDF 5 min (1).xlsx

Please upload NKE 5-min Excel file

NKE 5 min.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 25330

bytes, last modified: n/a - 100% done

Saving NKE 5 min.xlsx to NKE 5 min (1).xlsx

Please upload AMZN 30-min Excel file

AMZN 30 min.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) -

11054 bytes, last modified: n/a - 100% done

Saving AMZN 30 min.xlsx to AMZN 30 min (1).xlsx

Please upload KO 30-min Excel file

KO 30 min.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 11099

bytes, last modified: n/a - 100% done

Saving KO 30 min.xlsx to KO 30 min (1).xlsx

Please upload BYDDF 30-min Excel file

BYDDF 30 min.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) -

9693 bytes, last modified: n/a - 100% done

Saving BYDDF 30 min.xlsx to BYDDF 30 min (1).xlsx

Please upload NKE 30-min Excel file

NKE 30 min.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) -

11134 bytes, last modified: n/a - 100% done

Saving NKE 30 min.xlsx to NKE 30 min (1).xlsx

AMZN 5-min DataFrame columns: ['Date', 'Close']

KO 5-min DataFrame columns: ['Date', 'Close']

BYDDF 5-min DataFrame columns: ['Date', 'Close']

NKE 5-min DataFrame columns: ['Date', 'Close']

AMZN 30-min DataFrame columns: ['Date', 'Close']

KO 30-min DataFrame columns: ['Date', 'Close']

BYDDF 30-min DataFrame columns: ['Date', 'Close']

NKE 30-min DataFrame columns: ['Date', 'Close']

Please enter the column name containing closing prices: Close

Checking AMZN 5-min for invalid data:

- NaN values: 0
- Inf values: 0
- Non-numeric values: 0
- Length after cleaning: 943

Checking KO 5-min for invalid data:

- NaN values: 7
- Inf values: 0
- Non-numeric values: 0
- Length after cleaning: 942

Checking BYDDF 5-min for invalid data:

- NaN values: 1
- Inf values: 0
- Non-numeric values: 0
- Length after cleaning: 377

Checking NKE 5-min for invalid data:

- NaN values: 10
- Inf values: 0
- Non-numeric values: 0

- Length after cleaning: 944

Checking AMZN 30-min for invalid data:

- NaN values: 0
- Inf values: 0
- Non-numeric values: 0
- Length after cleaning: 130

Checking KO 30-min for invalid data:

- NaN values: 4
- Inf values: 0
- Non-numeric values: 0
- Length after cleaning: 130

Checking BYDDF 30-min for invalid data:

- NaN values: 0
- Inf values: 0
- Non-numeric values: 0
- Length after cleaning: 54

Checking NKE 30-min for invalid data:

- NaN values: 8
- Inf values: 0
- Non-numeric values: 0
- Length after cleaning: 130

Minimum length across all series: 54

Stationarity Tests for AMZN 5-min:

ADF Test:

ADF Statistic: -1.1414

p-value: 0.6983

Critical Values: {'1%': np.float64(-3.5656240522121956), '5%': np.float64(-2.920142229157715), '10%': np.float64(-2.598014675124952)}

Interpretation:

- p-value $\geq$ 0.05: Fail to reject null - AMZN 5-min may be non-stationary

KPSS Test:

KPSS Statistic: 0.7322

p-value: 0.0106

Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Interpretation:

- p-value < 0.05: Reject null hypothesis - AMZN 5-min is non-stationary

Stationarity Tests for KO 5-min:

ADF Test:

ADF Statistic: -2.9476

p-value: 0.0401

Critical Values: {'1%': np.float64(-3.560242358792829), '5%': np.float64(-2.9178502070837), '10%': np.float64(-2.5967964150943397)}

Interpretation:

- p-value < 0.05: Reject null hypothesis - KO 5-min is stationary

KPSS Test:

KPSS Statistic: 0.1044

p-value: 0.1000

Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Interpretation:

- p-value >= 0.05: Fail to reject null - KO 5-min may be stationary

Stationarity Tests for BYDDF 5-min:

ADF Test:

ADF Statistic: -2.6797

p-value: 0.0776

Critical Values: {'1%': np.float64(-3.562878534649522), '5%': np.float64(-2.918973284023669), '10%': np.float64(-2.597393446745562)}

Interpretation:

- p-value >= 0.05: Fail to reject null - BYDDF 5-min may be non-stationary

KPSS Test:

KPSS Statistic: 0.9310

p-value: 0.0100

Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Interpretation:

- p-value < 0.05: Reject null hypothesis - BYDDF 5-min is non-stationary

Stationarity Tests for NKE 5-min:

ADF Test:

ADF Statistic: -0.0785

p-value: 0.9515

Critical Values: {'1%': np.float64(-3.562878534649522), '5%': np.float64(-2.918973284023669), '10%': np.float64(-2.597393446745562)}

Interpretation:

- p-value >= 0.05: Fail to reject null - NKE 5-min may be non-stationary

KPSS Test:

KPSS Statistic: 1.0108

p-value: 0.0100

Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Interpretation:

- p-value < 0.05: Reject null hypothesis - NKE 5-min is non-stationary

Stationarity Tests for AMZN 30-min:

ADF Test:

ADF Statistic: -1.7445

p-value: 0.4083

Critical Values: {'1%': np.float64(-3.560242358792829), '5%': np.float64(-2.9178502070837), '10%': np.float64(-2.5967964150943397)}

Interpretation:

- p-value >= 0.05: Fail to reject null - AMZN 30-min may be non-stationary

KPSS Test:

KPSS Statistic: 0.7006

p-value: 0.0135

Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Interpretation:

- p-value < 0.05: Reject null hypothesis - AMZN 30-min is non-stationary

Stationarity Tests for KO 30-min:

ADF Test:

ADF Statistic: -4.6524

p-value: 0.0001

Critical Values: {'1%': np.float64(-3.560242358792829), '5%': np.float64(-2.9178502070837), '10%': np.float64(-2.5967964150943397)}

Interpretation:

- p-value < 0.05: Reject null hypothesis - KO 30-min is stationary

KPSS Test:

KPSS Statistic: 0.5853

p-value: 0.0240

Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Interpretation:

- p-value < 0.05: Reject null hypothesis - KO 30-min is non-stationary

Stationarity Tests for BYDDF 30-min:

ADF Test:

ADF Statistic: -0.7179

p-value: 0.8421

Critical Values: {'1%': np.float64(-3.560242358792829), '5%': np.float64(-2.9178502070837), '10%': np.float64(-2.5967964150943397)}

Interpretation:

- p-value >= 0.05: Fail to reject null - BYDDF 30-min may be non-stationary

KPSS Test:

KPSS Statistic: 1.0788

p-value: 0.0100

Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Interpretation:

- p-value < 0.05: Reject null hypothesis - BYDDF 30-min is non-stationary

Stationarity Tests for NKE 30-min:

ADF Test:

ADF Statistic: -1.2874

p-value: 0.6349

Critical Values: {'1%': np.float64(-3.560242358792829), '5%': np.float64(-2.9178502070837), '10%': np.float64(-2.5967964150943397)}

Interpretation:

- p-value >= 0.05: Fail to reject null - NKE 30-min may be non-stationary

KPSS Test:

KPSS Statistic: 0.8471

p-value: 0.0100

Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Interpretation:

- p-value < 0.05: Reject null hypothesis - NKE 30-min is non-stationary

Johansen Cointegration Test between AMZN 5-min and KO 5-min:

Trace Statistic:

Test Statistic: [9.64119682 2.45185926]

Critical Values (90%, 95%, 99%): [[13.4294 15.4943 19.9349]

[2.7055 3.8415 6.6349]]

Interpretation:

- Rank 0: No evidence of cointegration at 95% level
- Rank 1: No evidence of cointegration at 95% level

Johansen Cointegration Test between AMZN 5-min and BYDDF 5-min:

Trace Statistic:

Test Statistic: [27.31826999 2.95896333]

Critical Values (90%, 95%, 99%): [[13.4294 15.4943 19.9349]

[2.7055 3.8415 6.6349]]

Interpretation:

- Rank 0: Trace stat > 95% CV, evidence of cointegration
- Rank 1: No evidence of cointegration at 95% level

Johansen Cointegration Test between AMZN 5-min and NKE 5-min:

- Error: SVD did not converge. Consider differencing the series or checking data quality.

Johansen Cointegration Test between KO 5-min and BYDDF 5-min:

Trace Statistic:

Test Statistic: [11.30557265 3.49324557]

Critical Values (90%, 95%, 99%): [[13.4294 15.4943 19.9349]

[2.7055 3.8415 6.6349]]

Interpretation:

- Rank 0: No evidence of cointegration at 95% level
- Rank 1: No evidence of cointegration at 95% level

Johansen Cointegration Test between KO 5-min and NKE 5-min:

- Error: SVD did not converge. Consider differencing the series or checking data quality.

Johansen Cointegration Test between BYDDF 5-min and NKE 5-min:

- Error: SVD did not converge. Consider differencing the series or checking data quality.

Johansen Cointegration Test between AMZN 30-min and KO 30-min:

- Error: SVD did not converge. Consider differencing the series or checking data quality.

Johansen Cointegration Test between AMZN 30-min and BYDDF 30-min:

Trace Statistic:

Test Statistic: [5.50055161 0.28595805]

Critical Values (90%, 95%, 99%): [[13.4294 15.4943 19.9349]

[2.7055 3.8415 6.6349]]

Interpretation:

- Rank 0: No evidence of cointegration at 95% level

- Rank 1: No evidence of cointegration at 95% level

Johansen Cointegration Test between AMZN 30-min and NKE 30-min:

- Error: SVD did not converge. Consider differencing the series or checking data quality.

Johansen Cointegration Test between KO 30-min and BYDDF 30-min:

- Error: SVD did not converge. Consider differencing the series or checking data quality.

Johansen Cointegration Test between KO 30-min and NKE 30-min:

- Error: SVD did not converge. Consider differencing the series or checking data quality.

Johansen Cointegration Test between BYDDF 30-min and NKE 30-min:

- Error: SVD did not converge. Consider differencing the series or checking data quality.

Fitting Time Series Models for AMZN 5-min:

AR Model:

SARIMAX Results

=====

Dep. Variable: Close No. Observations: 54

Model: ARIMA(1, 0, 0) Log Likelihood -25.168

Date: Fri, 02 May 2025 AIC 56.337

Time: 23:15:07 BIC 62.304

Sample: 0 HQIC 58.638

- 54

Covariance Type: opg

=====

	coef	std err	z	P> z	[0.025	0.975]
const	190.1123	0.157	1212.955	0.000	189.805	190.419
ar.L1	0.5797	0.068	8.493	0.000	0.446	0.713
sigma2	0.1476	0.013	11.615	0.000	0.123	0.172

=====

=====

Ljung-Box (L1) (Q): 1.73 Jarque-Bera (JB): 405.14

Prob(Q): 0.19 Prob(JB): 0.00

Heteroskedasticity (H): 14.01 Skew: -1.97

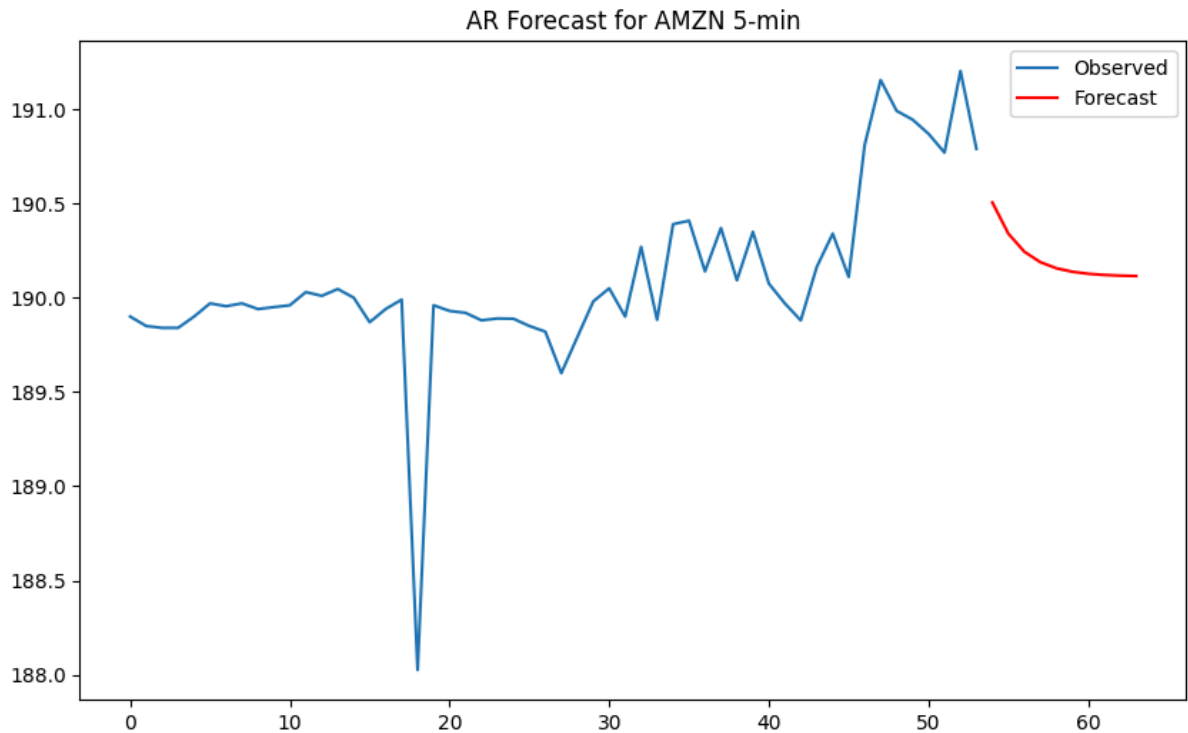
Prob(H) (two-sided): 0.00 Kurtosis: 15.83

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARMA Model:

SARIMAX Results

```
=====
Dep. Variable:      Close  No. Observations:      54
Model:              ARIMA(1, 0, 1)  Log Likelihood      -21.005
Date:              Fri, 02 May 2025  AIC              50.010
Time:              23:15:08  BIC              57.966
Sample:            0  HQIC              53.078
                    - 54
```

Covariance Type: opg

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const    190.2048    0.408  466.220   0.000   189.405   191.004
ar.L1     0.9431    0.131   7.200   0.000    0.686    1.200
ma.L1    -0.5973    0.187  -3.201   0.001   -0.963   -0.232
sigma2     0.1252    0.012  10.706   0.000    0.102    0.148
=====
```

```
=====
Ljung-Box (L1) (Q):      0.07  Jarque-Bera (JB):      700.68
Prob(Q):                 0.79  Prob(JB):           0.00
```

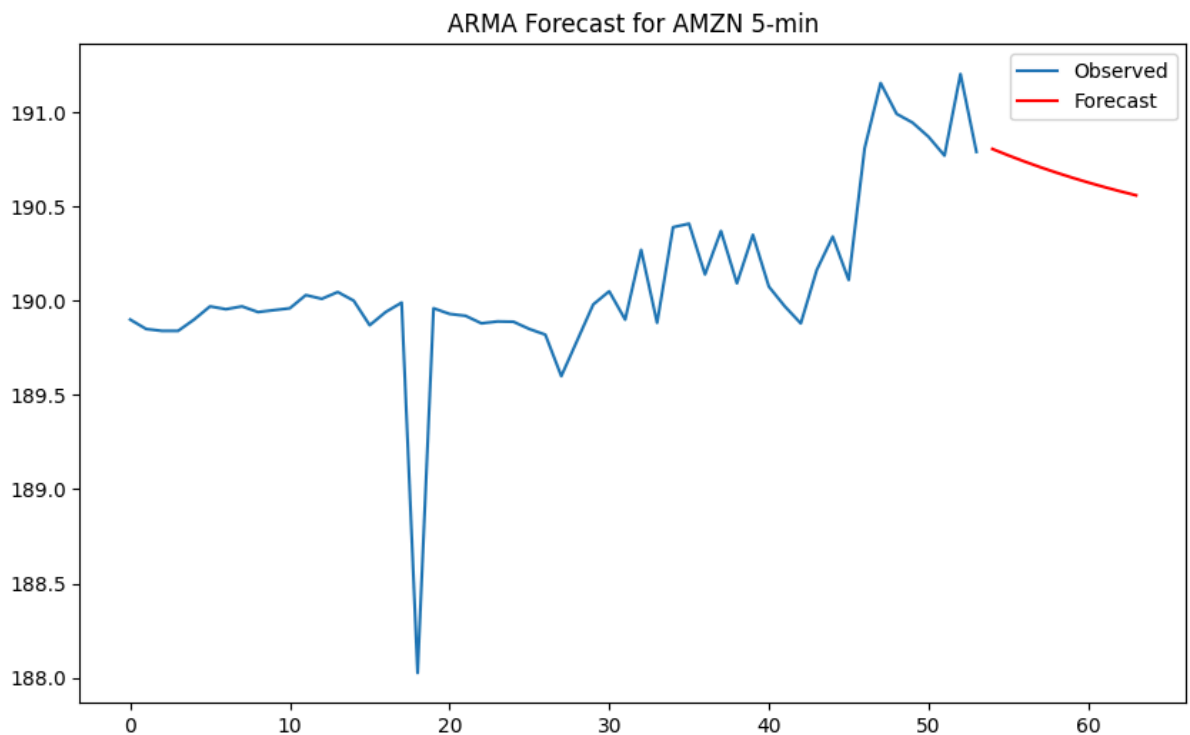
Heteroskedasticity (H):	13.99	Skew:	-2.89
Prob(H) (two-sided):	0.00	Kurtosis:	19.67

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARIMA Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
Model:	ARIMA(1, 1, 1)	Log Likelihood	-20.854
Date:	Fri, 02 May 2025	AIC	47.707
Time:	23:15:09	BIC	53.618
Sample:	0	HQIC	49.980
	- 54		

Covariance Type: opg

=====

coef	std err	z	P> z	[0.025	0.975]
------	---------	---	------	--------	--------

ar.L1	-0.0580	0.192	-0.303	0.762	-0.434	0.318
ma.L1	-0.6122	0.186	-3.299	0.001	-0.976	-0.248
sigma2	0.1273	0.009	14.609	0.000	0.110	0.144

=====

=====

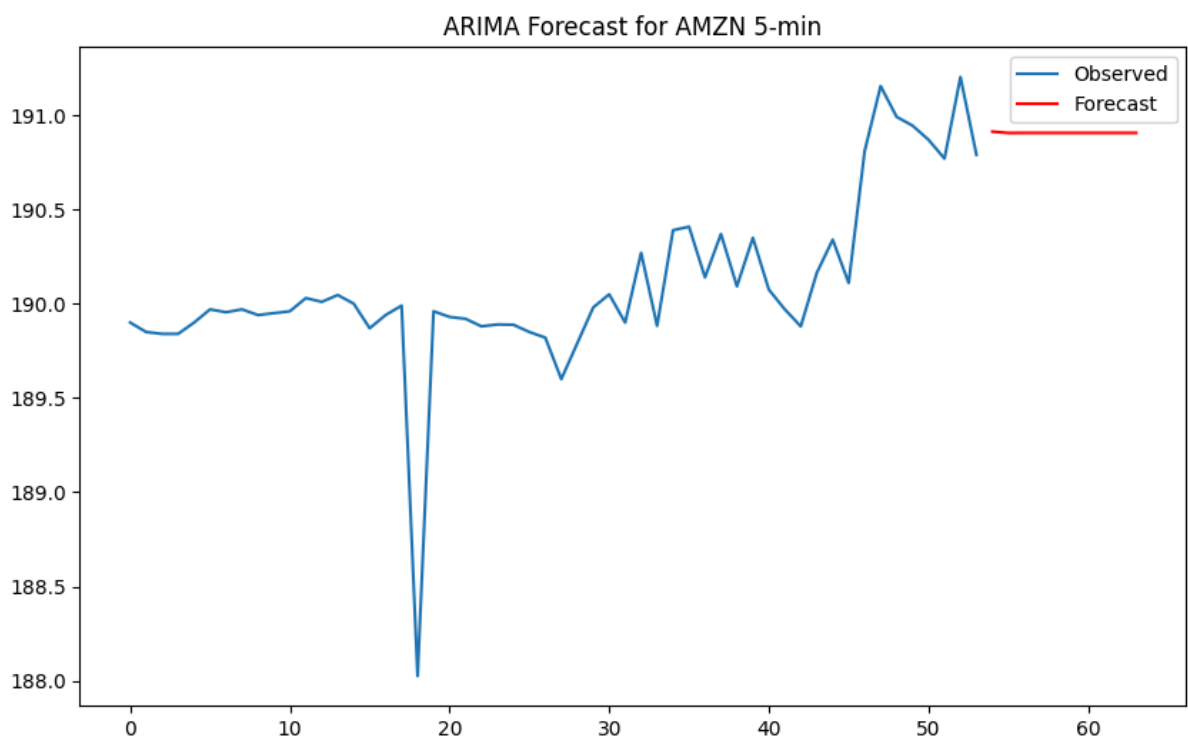
Ljung-Box (L1) (Q):	0.08	Jarque-Bera (JB):	707.15
Prob(Q):	0.78	Prob(JB):	0.00
Heteroskedasticity (H):	0.44	Skew:	-2.99
Prob(H) (two-sided):	0.09	Kurtosis:	19.87

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Fitting Time Series Models for KO 5-min:

AR Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
----------------	-------	-------------------	----

Model: ARIMA(1, 0, 0) Log Likelihood 103.773

Date: Fri, 02 May 2025 AIC -201.545

Time: 23:15:09 BIC -195.579

Sample: 0 HQIC -199.244

- 54

Covariance Type: opg

=====

	coef	std err	z	P> z	[0.025	0.975]
const	71.6903	0.015	4661.564	0.000	71.660	71.720
ar.L1	0.6608	0.123	5.381	0.000	0.420	0.902
sigma2	0.0012	0.000	5.176	0.000	0.001	0.002

=====

=====

Ljung-Box (L1) (Q): 0.73 Jarque-Bera (JB): 4.17

Prob(Q): 0.39 Prob(JB): 0.12

Heteroskedasticity (H): 1.19 Skew: 0.61

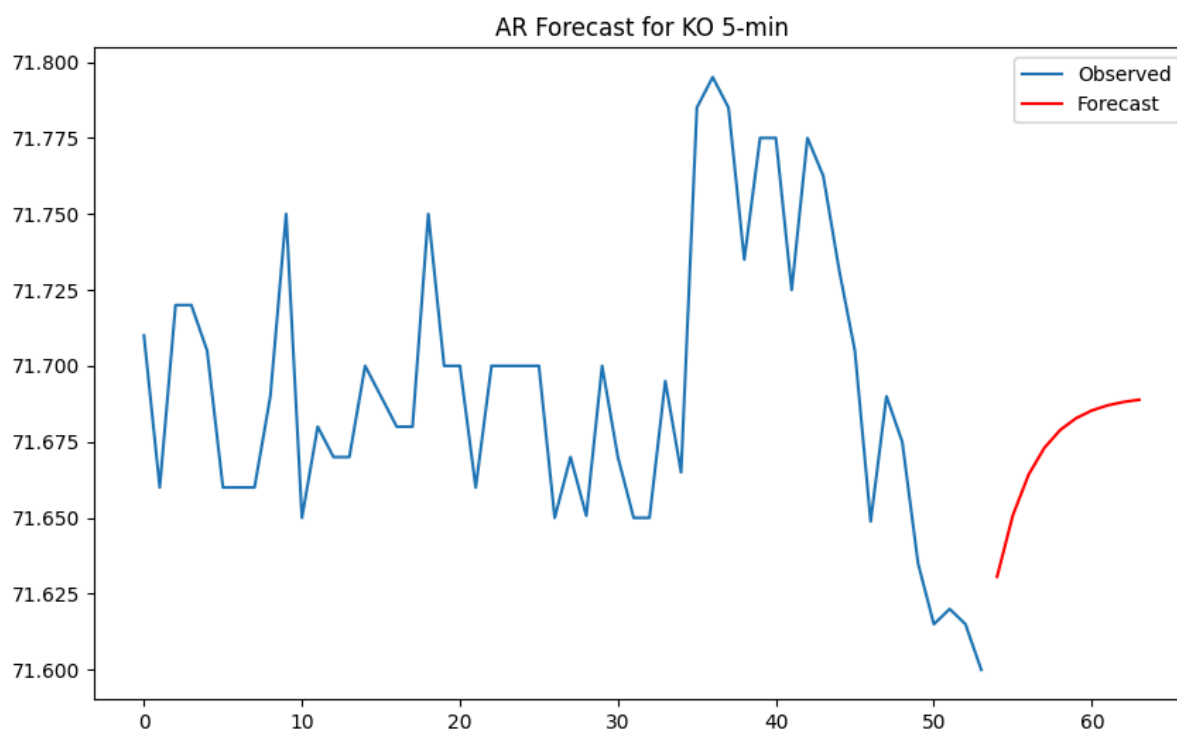
Prob(H) (two-sided): 0.72 Kurtosis: 3.62

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARMA Model:

SARIMAX Results

```
=====
Dep. Variable:      Close  No. Observations:      54
Model:              ARIMA(1, 0, 1)  Log Likelihood      104.726
Date:              Fri, 02 May 2025  AIC              -201.453
Time:              23:15:10  BIC              -193.497
Sample:            0  HQIC              -198.384
                    - 54
```

Covariance Type: opg

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const    71.6875   0.021 3460.107   0.000   71.647   71.728
ar.L1     0.8181   0.134   6.095   0.000   0.555   1.081
ma.L1    -0.2735   0.172  -1.591   0.112  -0.610   0.063
sigma2     0.0012   0.000   5.270   0.000   0.001   0.002
=====
```

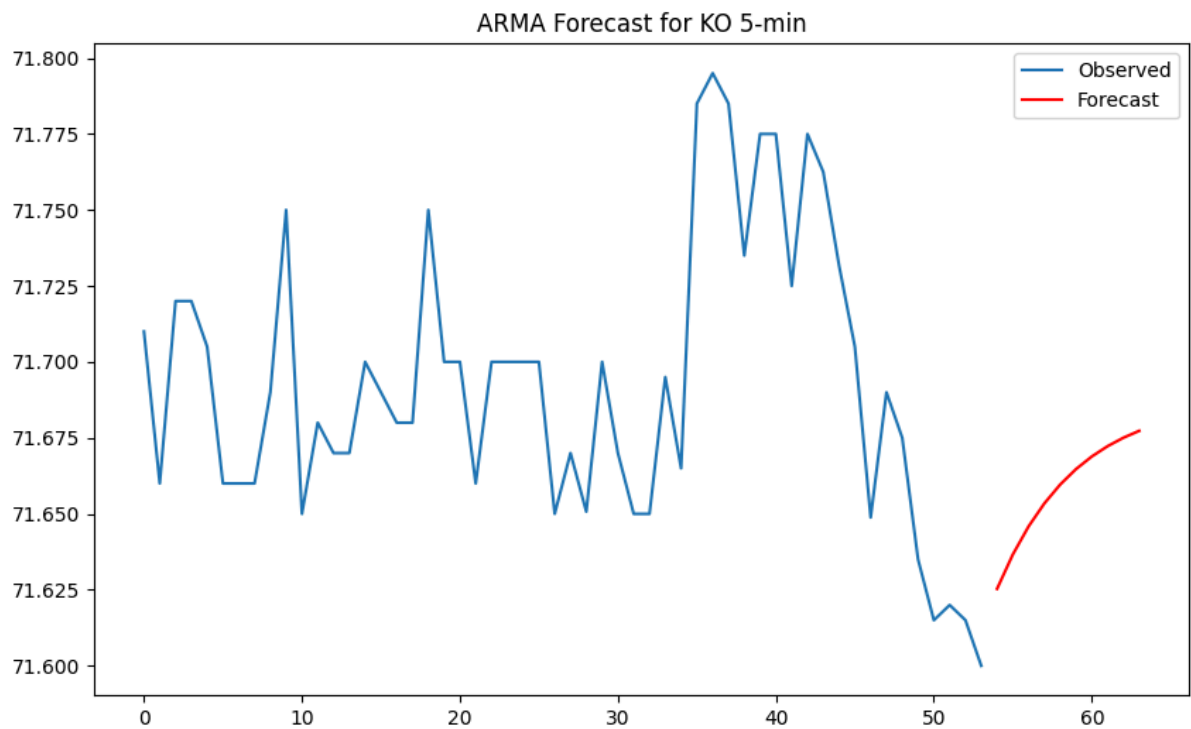
```
=====
Ljung-Box (L1) (Q):      0.00  Jarque-Bera (JB):      5.07
Prob(Q):                 0.99  Prob(JB):              0.08
```

Heteroskedasticity (H):	1.21	Skew:	0.67
Prob(H) (two-sided):	0.68	Kurtosis:	3.68

```
=====
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARIMA Model:

SARIMAX Results

```
=====
```

Dep. Variable:	Close	No. Observations:	54
Model:	ARIMA(1, 1, 1)	Log Likelihood	101.192
Date:	Fri, 02 May 2025	AIC	-196.383
Time:	23:15:10	BIC	-190.473
Sample:	0	HQIC	-194.110

- 54

Covariance Type: opg

```
=====
```

coef	std err	z	P> z	[0.025	0.975]
------	---------	---	------	--------	--------

```
-----
```

ar.L1	0.0565	0.455	0.124	0.901	-0.835	0.948
ma.L1	-0.4436	0.422	-1.051	0.293	-1.271	0.384
sigma2	0.0013	0.000	5.970	0.000	0.001	0.002

=====

=====

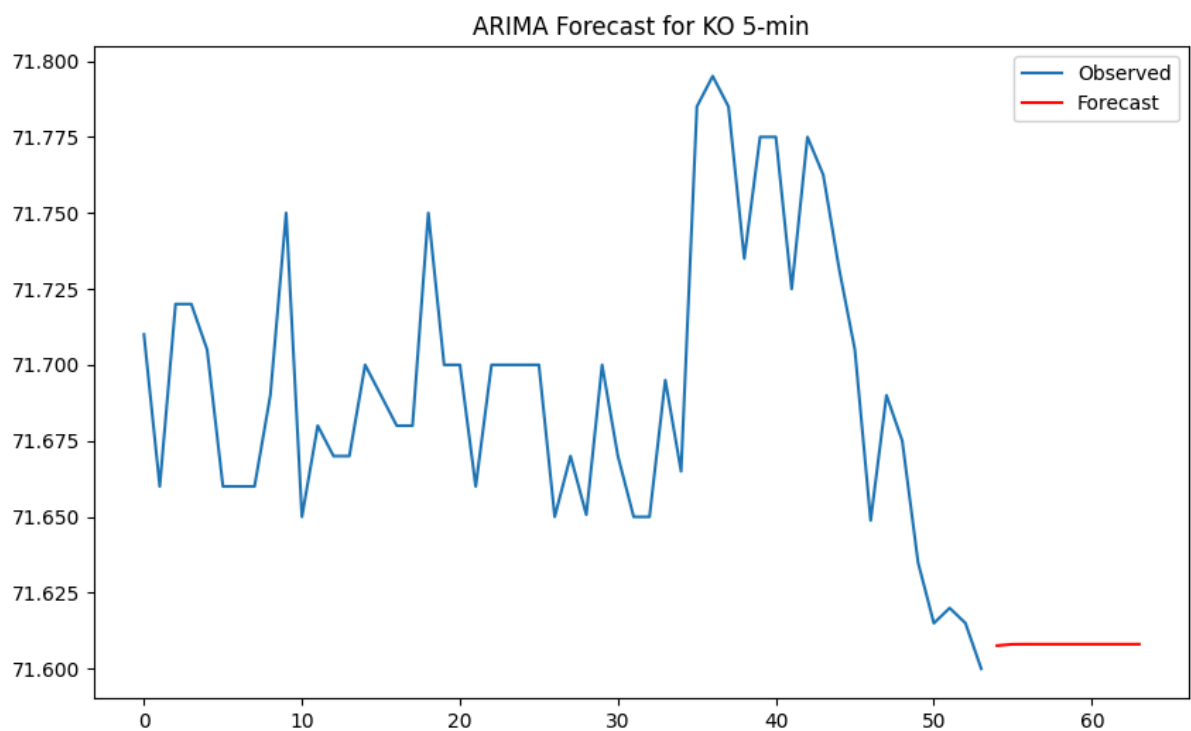
Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	7.04
Prob(Q):	0.96	Prob(JB):	0.03
Heteroskedasticity (H):	0.88	Skew:	0.70
Prob(H) (two-sided):	0.78	Kurtosis:	4.11

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Fitting Time Series Models for BYDDF 5-min:

AR Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
----------------	-------	-------------------	----

Model: ARIMA(1, 0, 0) Log Likelihood 65.787

Date: Fri, 02 May 2025 AIC -125.574

Time: 23:15:10 BIC -119.607

Sample: 0 HQIC -123.272

- 54

Covariance Type: opg

=====

	coef	std err	z	P> z	[0.025	0.975]
const	49.7027	0.030	1678.122	0.000	49.645	49.761
ar.L1	0.6429	0.081	7.969	0.000	0.485	0.801
sigma2	0.0051	0.001	6.035	0.000	0.003	0.007

=====

=====

Ljung-Box (L1) (Q): 0.39 Jarque-Bera (JB): 2.79

Prob(Q): 0.53 Prob(JB): 0.25

Heteroskedasticity (H): 0.48 Skew: 0.29

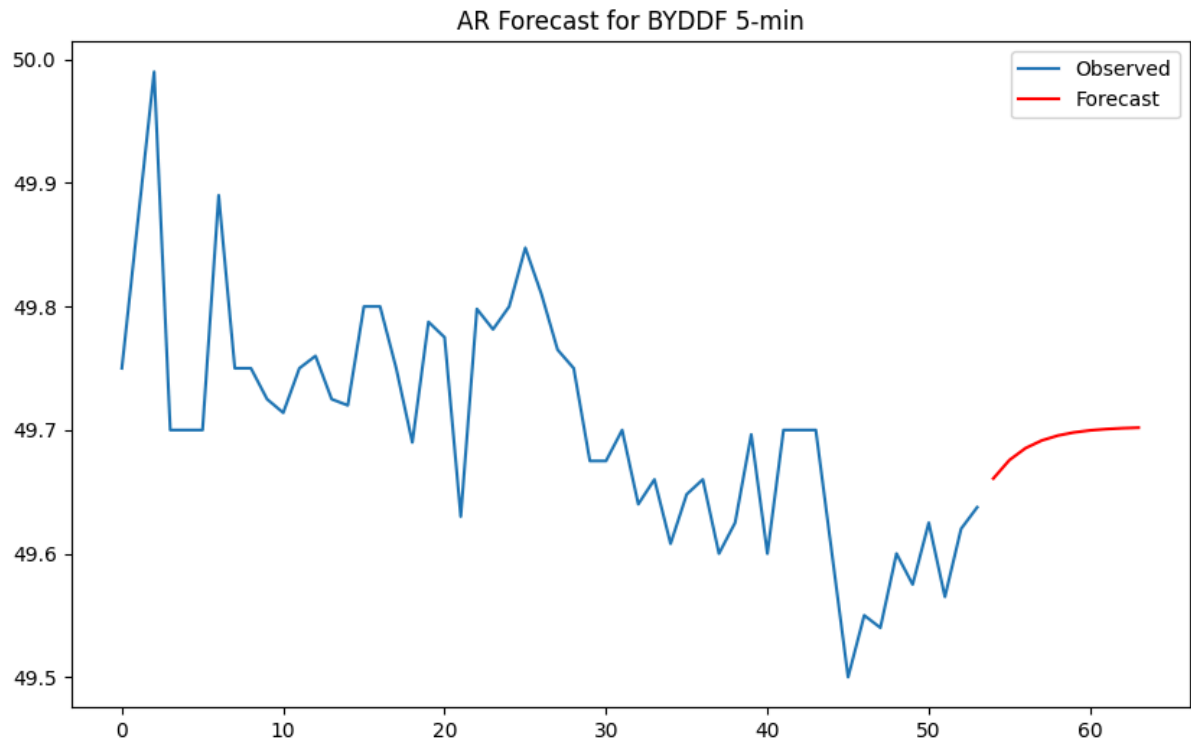
Prob(H) (two-sided): 0.12 Kurtosis: 3.95

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARMA Model:

SARIMAX Results

```
=====
Dep. Variable:      Close  No. Observations:      54
Model:              ARIMA(1, 0, 1)  Log Likelihood      67.645
Date:              Fri, 02 May 2025  AIC              -127.289
Time:              23:15:11  BIC              -119.333
Sample:            0  HQIC              -124.221
                    - 54
```

Covariance Type: opg

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const    49.7058    0.064  782.369   0.000   49.581   49.830
ar.L1     0.9355    0.085  11.051   0.000    0.770    1.101
ma.L1    -0.5831    0.167  -3.490   0.000   -0.911   -0.256
sigma2     0.0047    0.001   5.200   0.000    0.003    0.006
=====
```

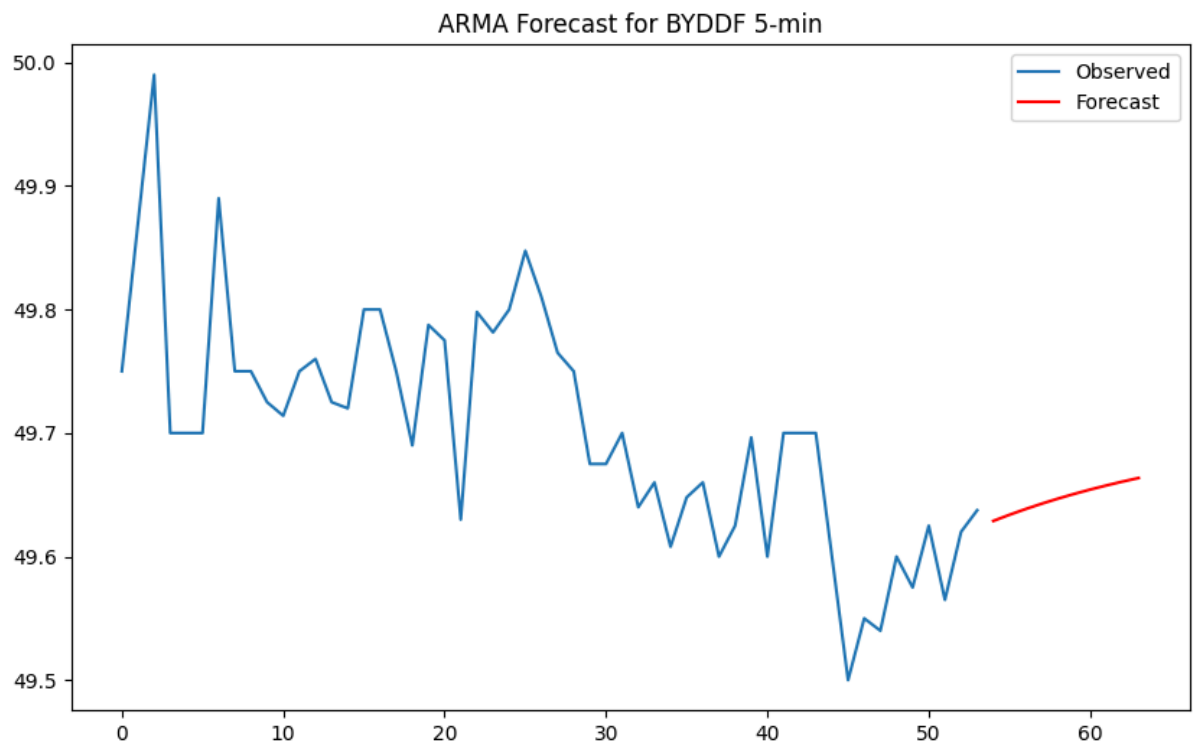
```
=====
Ljung-Box (L1) (Q):      0.66  Jarque-Bera (JB):      2.16
Prob(Q):                 0.42  Prob(JB):            0.34
```

Heteroskedasticity (H):	0.46	Skew:	0.32
Prob(H) (two-sided):	0.11	Kurtosis:	3.75

```
=====
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARIMA Model:

SARIMAX Results

```
=====
```

Dep. Variable:	Close	No. Observations:	54
Model:	ARIMA(1, 1, 1)	Log Likelihood	66.800
Date:	Fri, 02 May 2025	AIC	-127.600
Time:	23:15:11	BIC	-121.689
Sample:	0	HQIC	-125.327
	- 54		

Covariance Type: opg

```
=====
```

coef	std err	z	P> z	[0.025	0.975]
------	---------	---	------	--------	--------

```
-----
```

ar.L1	0.2325	0.188	1.239	0.215	-0.135	0.600
ma.L1	-0.7910	0.172	-4.589	0.000	-1.129	-0.453
sigma2	0.0047	0.001	4.860	0.000	0.003	0.007

=====

=====

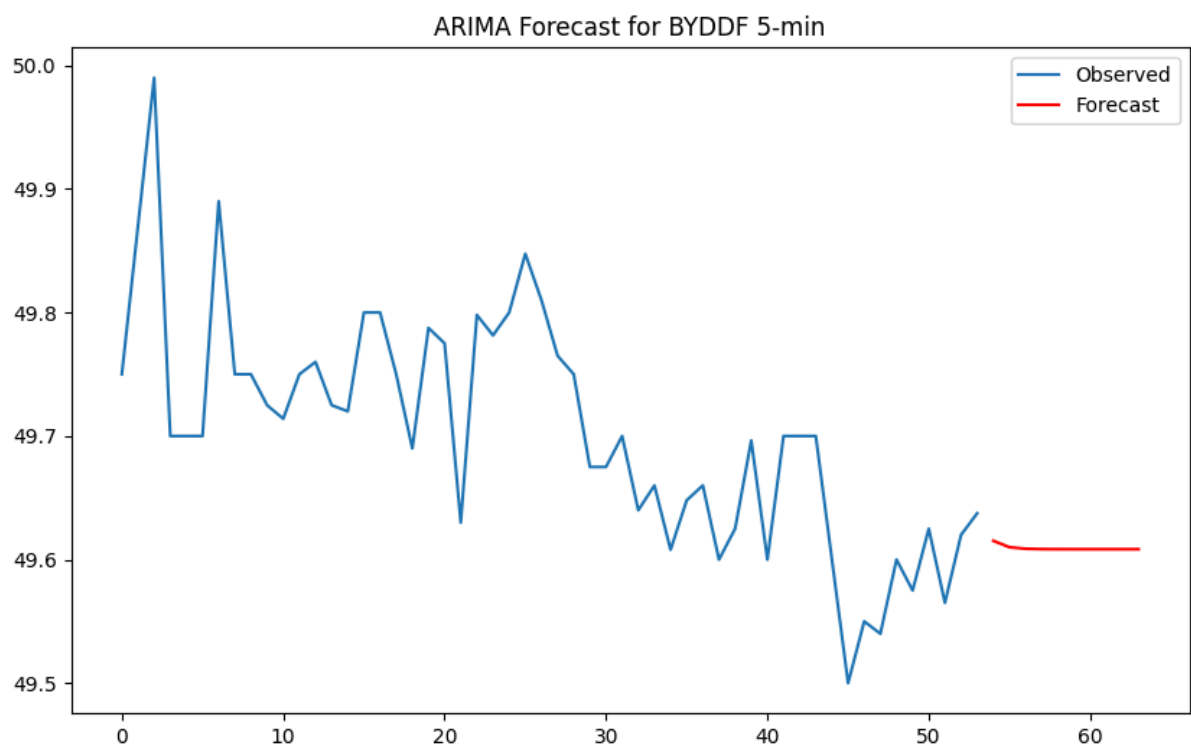
Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	0.41
Prob(Q):	0.94	Prob(JB):	0.81
Heteroskedasticity (H):	0.45	Skew:	0.04
Prob(H) (two-sided):	0.10	Kurtosis:	3.43

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Fitting Time Series Models for NKE 5-min:

AR Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
----------------	-------	-------------------	----

Model: ARIMA(1, 0, 0) Log Likelihood 65.620

Date: Fri, 02 May 2025 AIC -125.239

Time: 23:15:11 BIC -119.272

Sample: 0 HQIC -122.938

- 54

Covariance Type: opg

=====

	coef	std err	z	P> z	[0.025	0.975]
const	58.5094	0.181	322.798	0.000	58.154	58.865
ar.L1	0.9616	0.050	19.254	0.000	0.864	1.059
sigma2	0.0049	0.001	7.333	0.000	0.004	0.006

=====

=====

Ljung-Box (L1) (Q): 4.16 Jarque-Bera (JB): 10.52

Prob(Q): 0.04 Prob(JB): 0.01

Heteroskedasticity (H): 2.38 Skew: -0.17

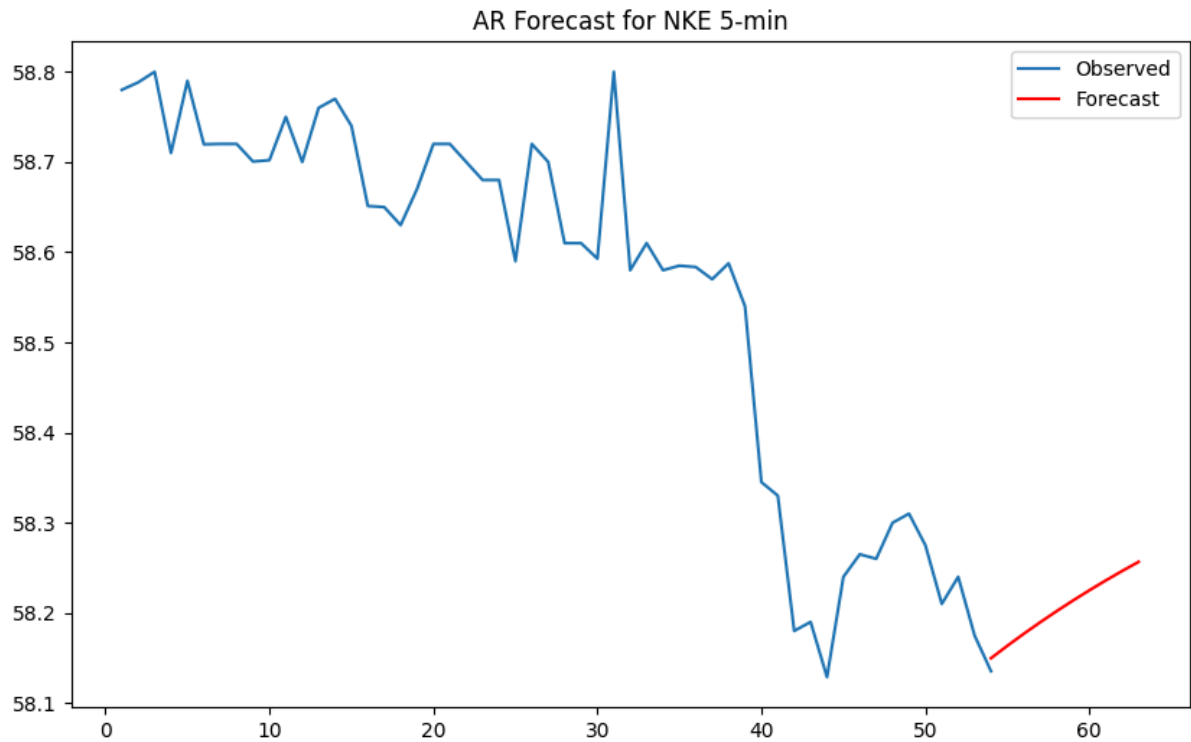
Prob(H) (two-sided): 0.07 Kurtosis: 5.13

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARMA Model:

SARIMAX Results

```
=====
Dep. Variable:      Close  No. Observations:      54
Model:              ARIMA(1, 0, 1)  Log Likelihood      67.120
Date:              Fri, 02 May 2025  AIC              -126.241
Time:              23:15:12  BIC              -118.285
Sample:            0  HQIC              -123.173
```

- 54

Covariance Type: opg

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const    58.4971    0.229  255.158   0.000   58.048   58.946
ar.L1     0.9803    0.048   20.233   0.000    0.885    1.075
ma.L1    -0.2157    0.127   -1.696   0.090   -0.465    0.034
sigma2     0.0046    0.001    7.106   0.000    0.003    0.006
=====
```

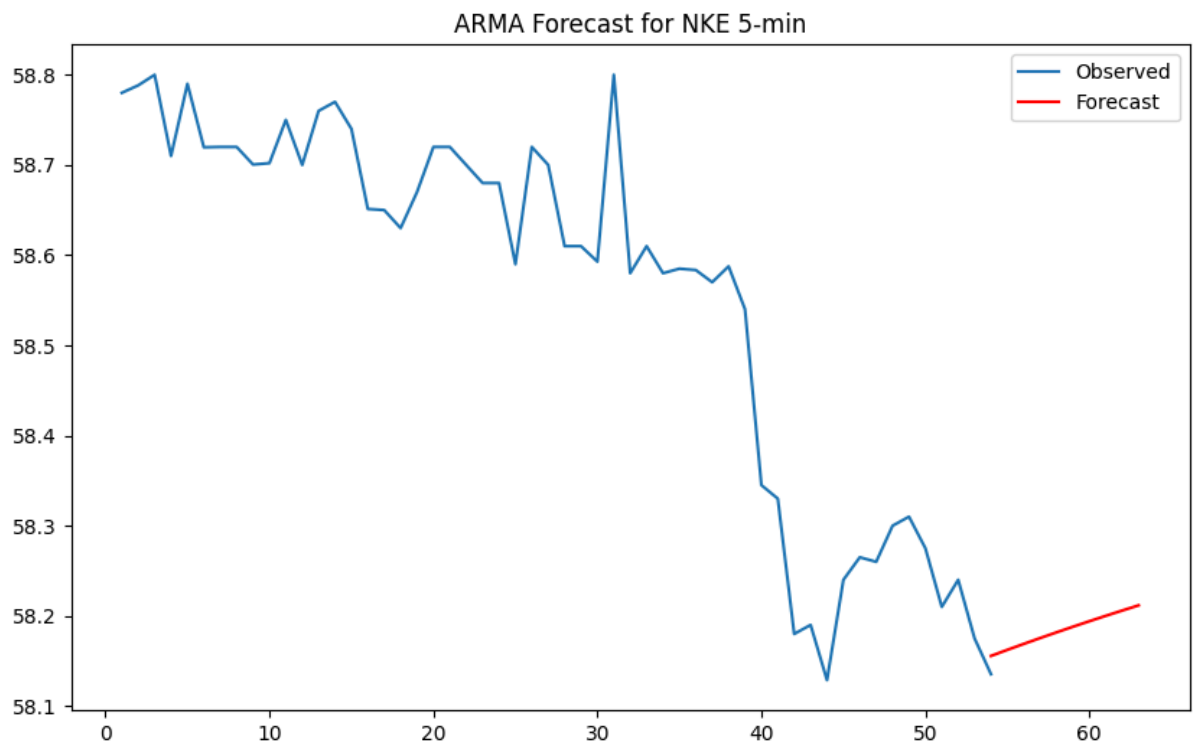
```
=====
Ljung-Box (L1) (Q):      0.29  Jarque-Bera (JB):      9.19
Prob(Q):      0.59  Prob(JB):      0.01
```

Heteroskedasticity (H):	3.00	Skew:	-0.09
Prob(H) (two-sided):	0.02	Kurtosis:	5.01

```
=====
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARIMA Model:

SARIMAX Results

```
=====
```

Dep. Variable:	Close	No. Observations:	54
Model:	ARIMA(1, 1, 1)	Log Likelihood	67.897
Date:	Fri, 02 May 2025	AIC	-129.795
Time:	23:15:12	BIC	-123.884
Sample:	0	HQIC	-127.522
	- 54		

Covariance Type: opg

```
=====
```

	coef	std err	z	P> z	[0.025	0.975]

ar.L1	-0.5668	0.478	-1.186	0.236	-1.504	0.370
ma.L1	0.3078	0.520	0.592	0.554	-0.712	1.327
sigma2	0.0045	0.001	6.986	0.000	0.003	0.006

=====

=====

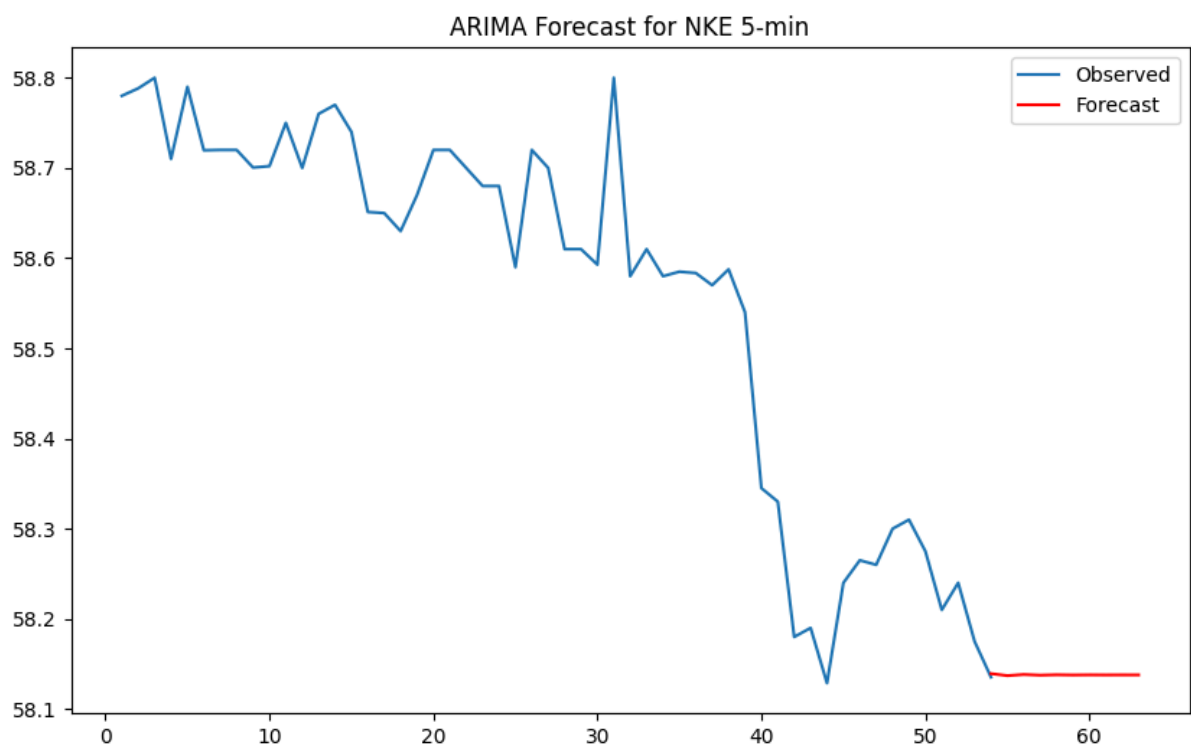
Ljung-Box (L1) (Q):	0.09	Jarque-Bera (JB):	10.07
Prob(Q):	0.76	Prob(JB):	0.01
Heteroskedasticity (H):	3.03	Skew:	-0.02
Prob(H) (two-sided):	0.02	Kurtosis:	5.13

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Fitting Time Series Models for AMZN 30-min:

AR Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
----------------	-------	-------------------	----

Model: ARIMA(1, 0, 0) Log Likelihood -82.616

Date: Fri, 02 May 2025 AIC 171.233

Time: 23:15:12 BIC 177.199

Sample: 0 HQIC 173.534

- 54

Covariance Type: opg

=====

	coef	std err	z	P> z	[0.025	0.975]
const	186.2833	1.645	113.211	0.000	183.058	189.508
ar.L1	0.9096	0.057	15.921	0.000	0.798	1.022
sigma2	1.2086	0.158	7.650	0.000	0.899	1.518

=====

=====

Ljung-Box (L1) (Q): 1.84 Jarque-Bera (JB): 61.55

Prob(Q): 0.17 Prob(JB): 0.00

Heteroskedasticity (H): 0.41 Skew: -0.90

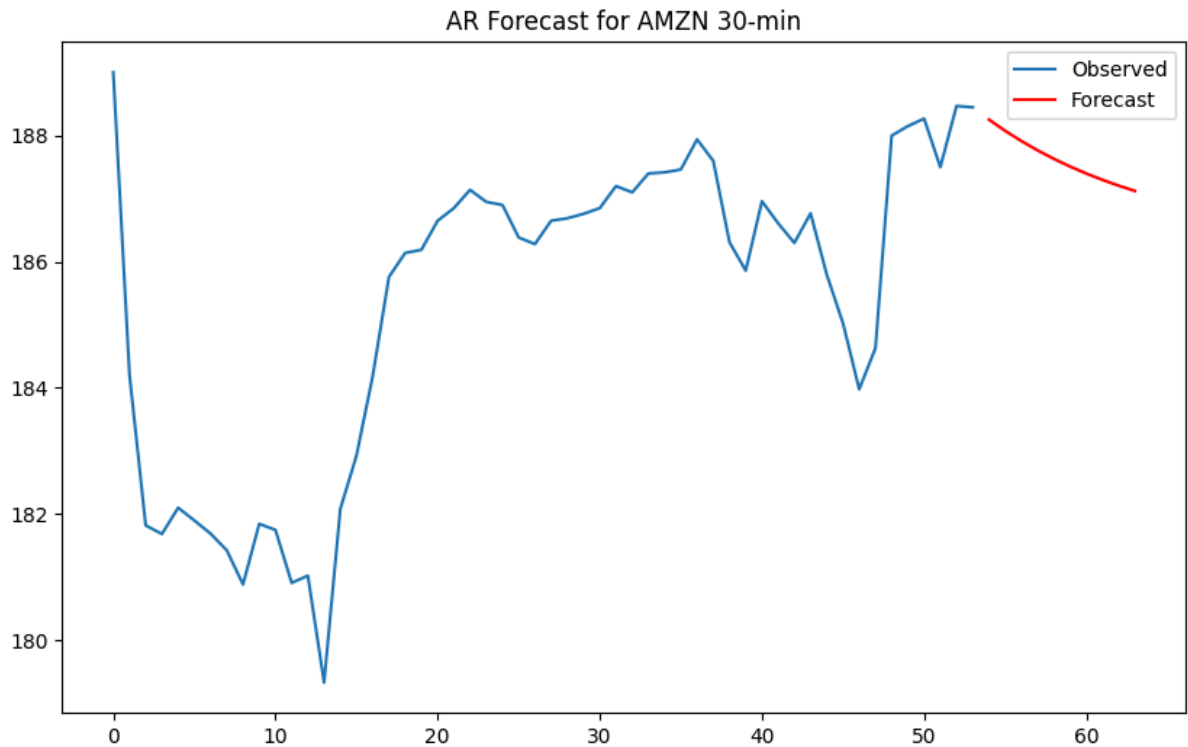
Prob(H) (two-sided): 0.07 Kurtosis: 7.91

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARMA Model:

SARIMAX Results

```
=====
Dep. Variable:      Close  No. Observations:      54
Model:              ARIMA(1, 0, 1)  Log Likelihood      -80.382
Date:              Fri, 02 May 2025  AIC              168.765
Time:              23:15:12  BIC              176.721
Sample:            0  HQIC              171.833
                    - 54
```

Covariance Type: opg

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const    186.1515    1.515  122.866   0.000   183.182   189.121
ar.L1     0.8648     0.079   10.885   0.000    0.709    1.021
ma.L1     0.3257     0.143    2.281   0.023    0.046    0.606
sigma2     1.1079     0.158    7.007   0.000    0.798    1.418
=====
```

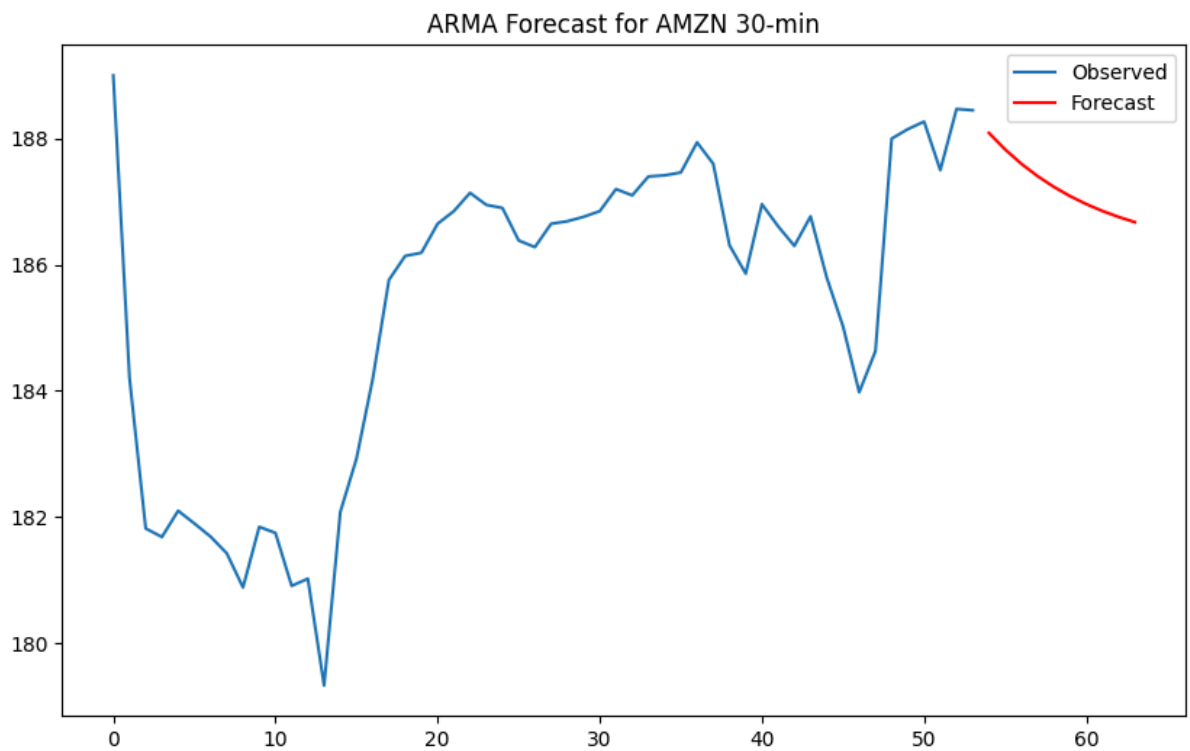
```
=====
Ljung-Box (L1) (Q):      0.57  Jarque-Bera (JB):      56.20
Prob(Q):      0.45  Prob(JB):      0.00
```

Heteroskedasticity (H):	0.43	Skew:	-0.66
Prob(H) (two-sided):	0.09	Kurtosis:	7.82

```
=====
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARIMA Model:

SARIMAX Results

```
=====
```

Dep. Variable:	Close	No. Observations:	54
Model:	ARIMA(1, 1, 1)	Log Likelihood	-79.501
Date:	Fri, 02 May 2025	AIC	165.002
Time:	23:15:13	BIC	170.913
Sample:	0	HQIC	167.275
	- 54		

Covariance Type: opg

```
=====
```

coef	std err	z	P> z	[0.025	0.975]
------	---------	---	------	--------	--------

```
-----
```

ar.L1	0.2694	0.577	0.467	0.641	-0.861	1.400
ma.L1	0.0580	0.588	0.099	0.921	-1.095	1.211
sigma2	1.1736	0.122	9.648	0.000	0.935	1.412

=====

=====

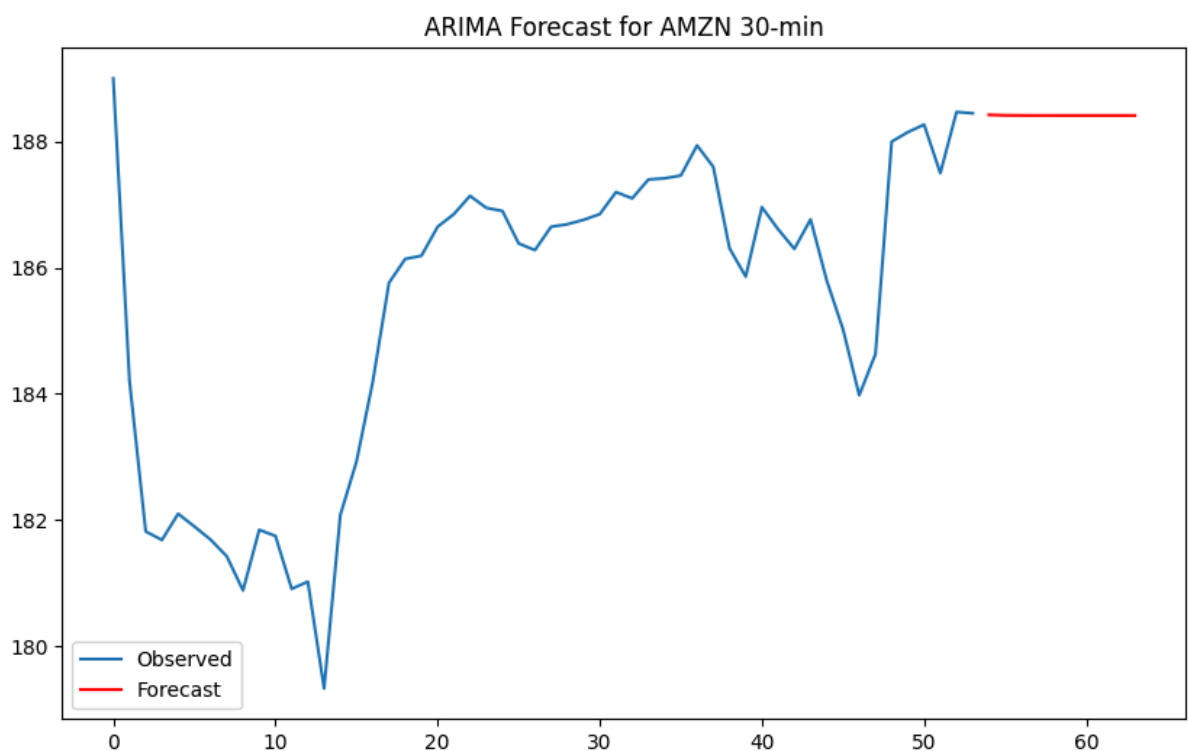
Ljung-Box (L1) (Q):	0.50	Jarque-Bera (JB):	84.57
Prob(Q):	0.48	Prob(JB):	0.00
Heteroskedasticity (H):	0.50	Skew:	-0.42
Prob(H) (two-sided):	0.15	Kurtosis:	9.13

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Fitting Time Series Models for KO 30-min:

AR Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
----------------	-------	-------------------	----

Model: ARIMA(1, 0, 0) Log Likelihood -4.077

Date: Fri, 02 May 2025 AIC 14.155

Time: 23:15:13 BIC 20.122

Sample: 0 HQIC 16.456

- 54

Covariance Type: opg

=====

	coef	std err	z	P> z	[0.025	0.975]
const	72.3076	0.087	829.747	0.000	72.137	72.478
ar.L1	0.3576	0.143	2.507	0.012	0.078	0.637
sigma2	0.0679	0.006	10.896	0.000	0.056	0.080

=====

=====

Ljung-Box (L1) (Q): 0.19 Jarque-Bera (JB): 218.65

Prob(Q): 0.66 Prob(JB): 0.00

Heteroskedasticity (H): 0.70 Skew: 1.53

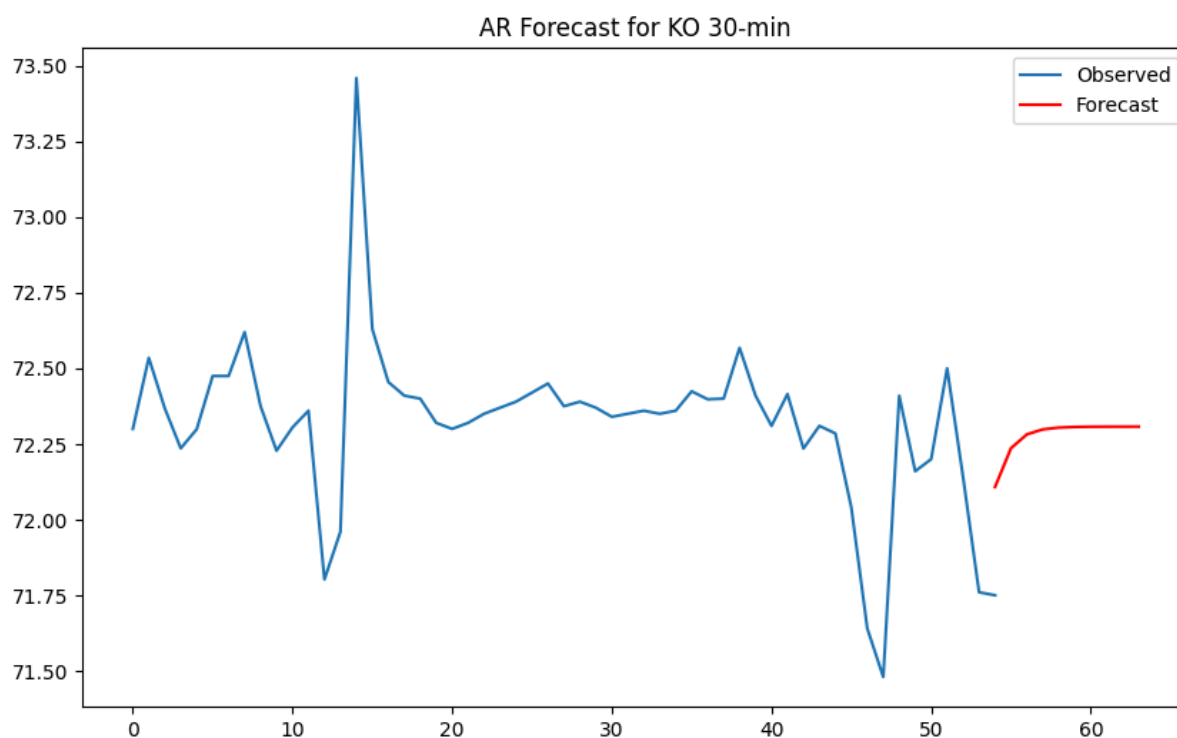
Prob(H) (two-sided): 0.46 Kurtosis: 12.37

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARMA Model:

SARIMAX Results

```
=====
Dep. Variable:      Close  No. Observations:      54
Model:              ARIMA(1, 0, 1)  Log Likelihood      -2.686
Date:              Fri, 02 May 2025  AIC              13.372
Time:              23:15:13  BIC              21.328
Sample:            0  HQIC              16.440
                    - 54
```

Covariance Type: opg

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const    72.3104   0.063 1146.498   0.000   72.187   72.434
ar.L1    -0.1853   0.296  -0.627   0.531  -0.765   0.394
ma.L1     0.6274   0.269   2.328   0.020   0.099   1.155
sigma2     0.0643   0.008   7.606   0.000   0.048   0.081
=====
```

```
=====
Ljung-Box (L1) (Q):      0.01  Jarque-Bera (JB):      115.86
Prob(Q):                  0.93  Prob(JB):            0.00
```

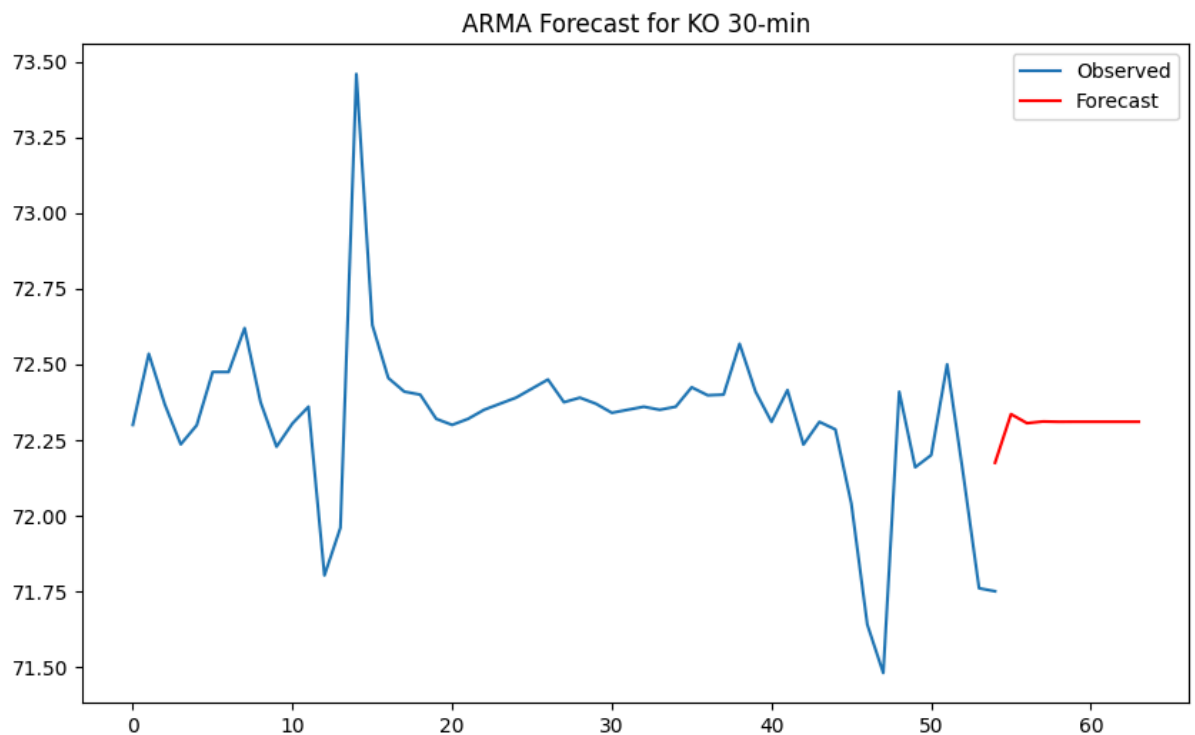
Heteroskedasticity (H):	0.74	Skew:	1.02
Prob(H) (two-sided):	0.54	Kurtosis:	9.88

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARIMA Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
Model:	ARIMA(1, 1, 1)	Log Likelihood	-5.183
Date:	Fri, 02 May 2025	AIC	16.365
Time:	23:15:14	BIC	22.276
Sample:	0	HQIC	18.638
	- 54		

Covariance Type: opg

=====

coef	std err	z	P> z	[0.025	0.975]
------	---------	---	------	--------	--------

ar.L1	0.2757	0.278	0.991	0.322	-0.270	0.821
ma.L1	-0.8995	0.161	-5.586	0.000	-1.215	-0.584
sigma2	0.0696	0.007	10.359	0.000	0.056	0.083

=====

=====

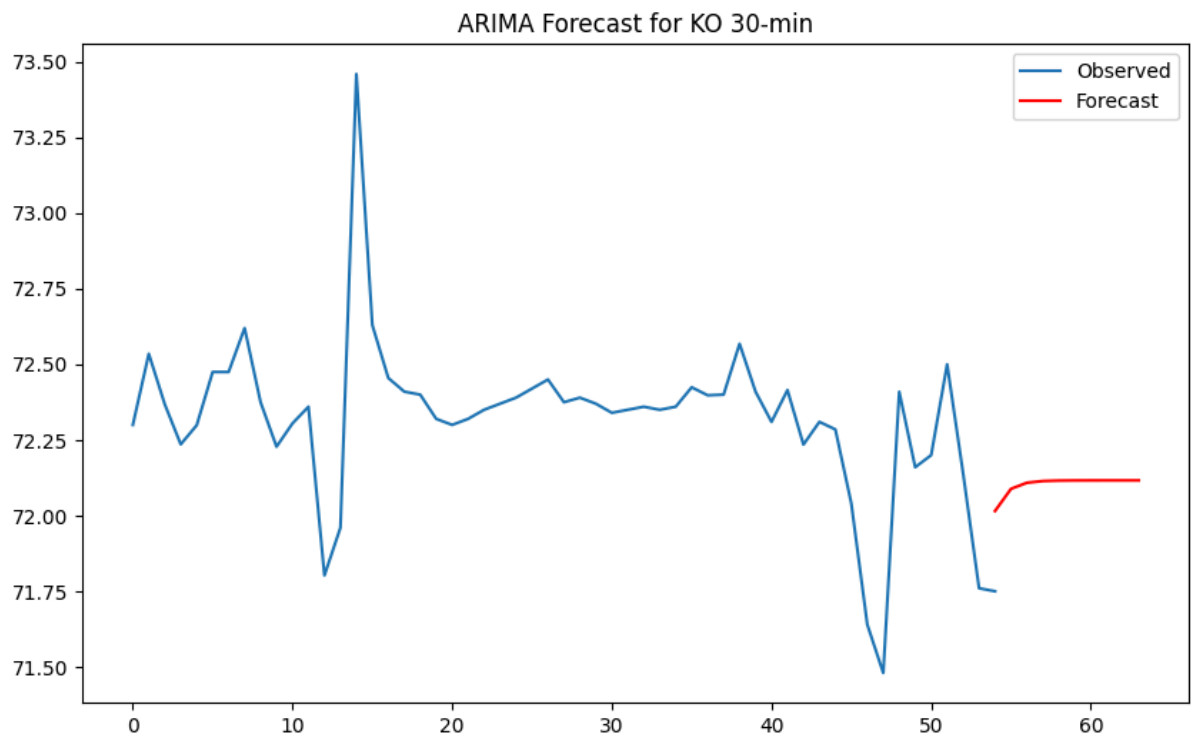
Ljung-Box (L1) (Q):	0.06	Jarque-Bera (JB):	275.97
Prob(Q):	0.81	Prob(JB):	0.00
Heteroskedasticity (H):	0.68	Skew:	1.92
Prob(H) (two-sided):	0.43	Kurtosis:	13.50

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Fitting Time Series Models for BYDDF 30-min:

AR Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
----------------	-------	-------------------	----

Model: ARIMA(1, 0, 0) Log Likelihood -31.917

Date: Fri, 02 May 2025 AIC 69.834

Time: 23:15:14 BIC 75.800

Sample: 0 HQIC 72.135

- 54

Covariance Type: opg

=====

	coef	std err	z	P> z	[0.025	0.975]
const	49.4259	1.718	28.776	0.000	46.059	52.792
ar.L1	0.9735	0.053	18.503	0.000	0.870	1.077
sigma2	0.1808	0.013	14.159	0.000	0.156	0.206

=====

=====

Ljung-Box (L1) (Q): 0.14 Jarque-Bera (JB): 906.17

Prob(Q): 0.71 Prob(JB): 0.00

Heteroskedasticity (H): 9.15 Skew: 3.93

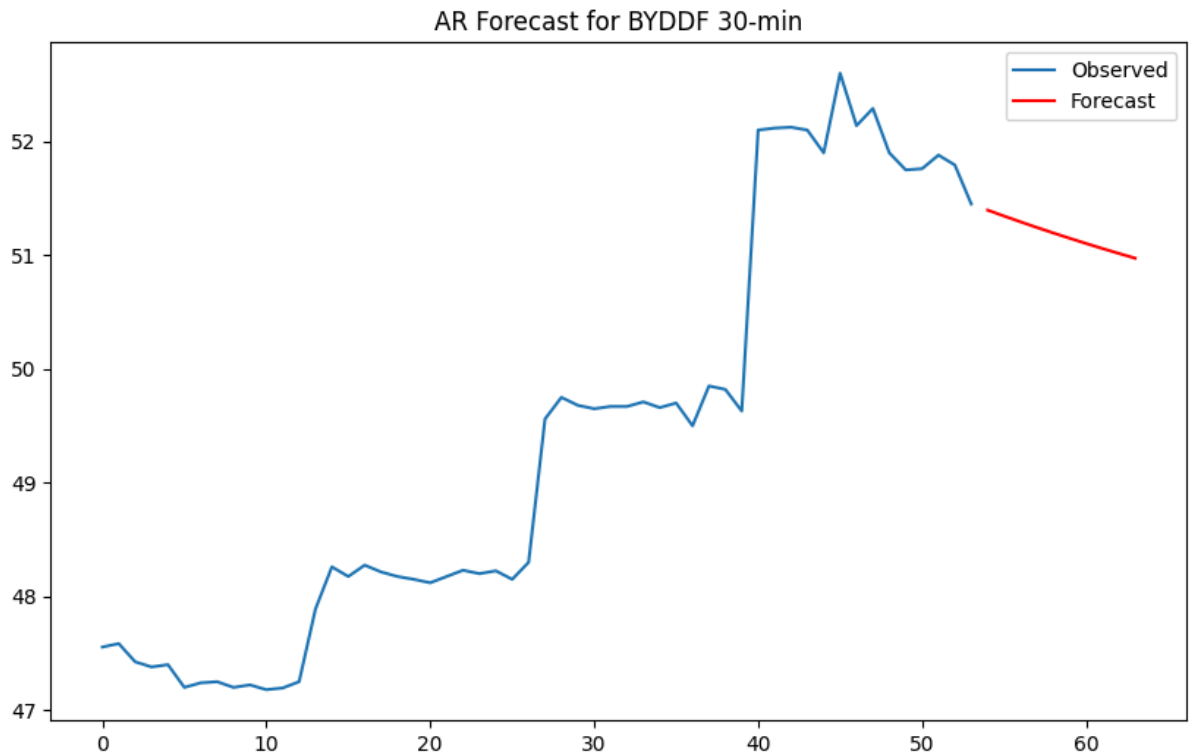
Prob(H) (two-sided): 0.00 Kurtosis: 21.47

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARMA Model:

SARIMAX Results

```
=====
Dep. Variable:      Close  No. Observations:      54
Model:              ARIMA(1, 0, 1)  Log Likelihood      -31.900
Date:              Fri, 02 May 2025  AIC              71.800
Time:              23:15:14  BIC              79.756
Sample:            0  HQIC              74.869
                    - 54
```

Covariance Type: opg

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const    49.4311    1.746   28.309   0.000    46.009    52.853
ar.L1     0.9749    0.051   18.953   0.000     0.874     1.076
ma.L1    -0.0252    0.291   -0.087   0.931    -0.595     0.545
sigma2     0.1807    0.015   12.015   0.000     0.151     0.210
=====
```

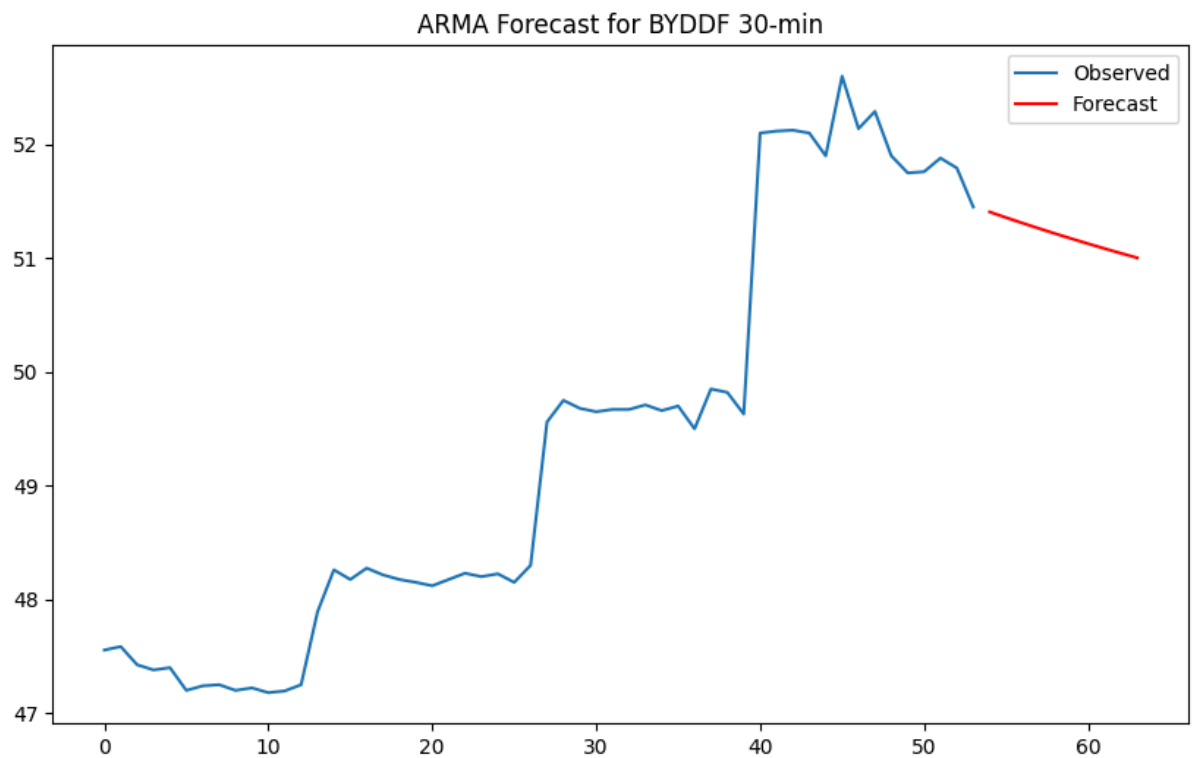
```
=====
Ljung-Box (L1) (Q):      0.04  Jarque-Bera (JB):      893.55
Prob(Q):                 0.83  Prob(JB):           0.00
```

Heteroskedasticity (H):	9.00	Skew:	3.91
Prob(H) (two-sided):	0.00	Kurtosis:	21.33

```
=====
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARIMA Model:

SARIMAX Results

```
=====
```

Dep. Variable:	Close	No. Observations:	54
Model:	ARIMA(1, 1, 1)	Log Likelihood	-30.161
Date:	Fri, 02 May 2025	AIC	66.322
Time:	23:15:15	BIC	72.233
Sample:	0	HQIC	68.595

- 54

Covariance Type: opg

```
=====
```

coef	std err	z	P> z	[0.025	0.975]
------	---------	---	------	--------	--------

```
-----
```

ar.L1	-0.0530	16.791	-0.003	0.997	-32.963	32.857
ma.L1	0.0118	16.871	0.001	0.999	-33.056	33.079
sigma2	0.1827	0.014	12.724	0.000	0.155	0.211

=====

=====

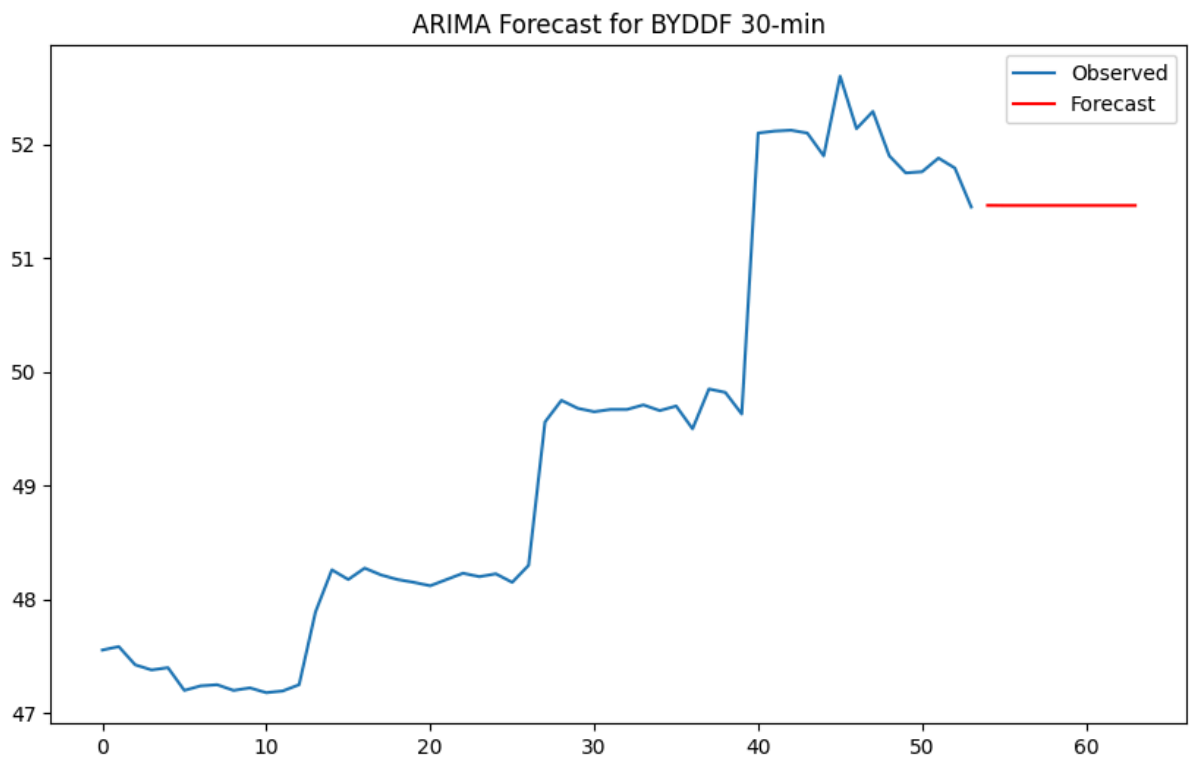
Ljung-Box (L1) (Q):	0.08	Jarque-Bera (JB):	849.01
Prob(Q):	0.78	Prob(JB):	0.00
Heteroskedasticity (H):	10.97	Skew:	3.88
Prob(H) (two-sided):	0.00	Kurtosis:	21.01

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Fitting Time Series Models for NKE 30-min:

AR Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
----------------	-------	-------------------	----

Model: ARIMA(1, 0, 0) Log Likelihood -2.502

Date: Fri, 02 May 2025 AIC 11.004

Time: 23:15:15 BIC 16.971

Sample: 0 HQIC 13.305

- 54

Covariance Type: opg

=====

	coef	std err	z	P> z	[0.025	0.975]

const	56.8938	0.464	122.573	0.000	55.984	57.804
ar.L1	0.9254	0.045	20.663	0.000	0.838	1.013
sigma2	0.0620	0.011	5.558	0.000	0.040	0.084

=====

=====

Ljung-Box (L1) (Q): 0.13 Jarque-Bera (JB): 3.90

Prob(Q): 0.72 Prob(JB): 0.14

Heteroskedasticity (H): 0.38 Skew: -0.57

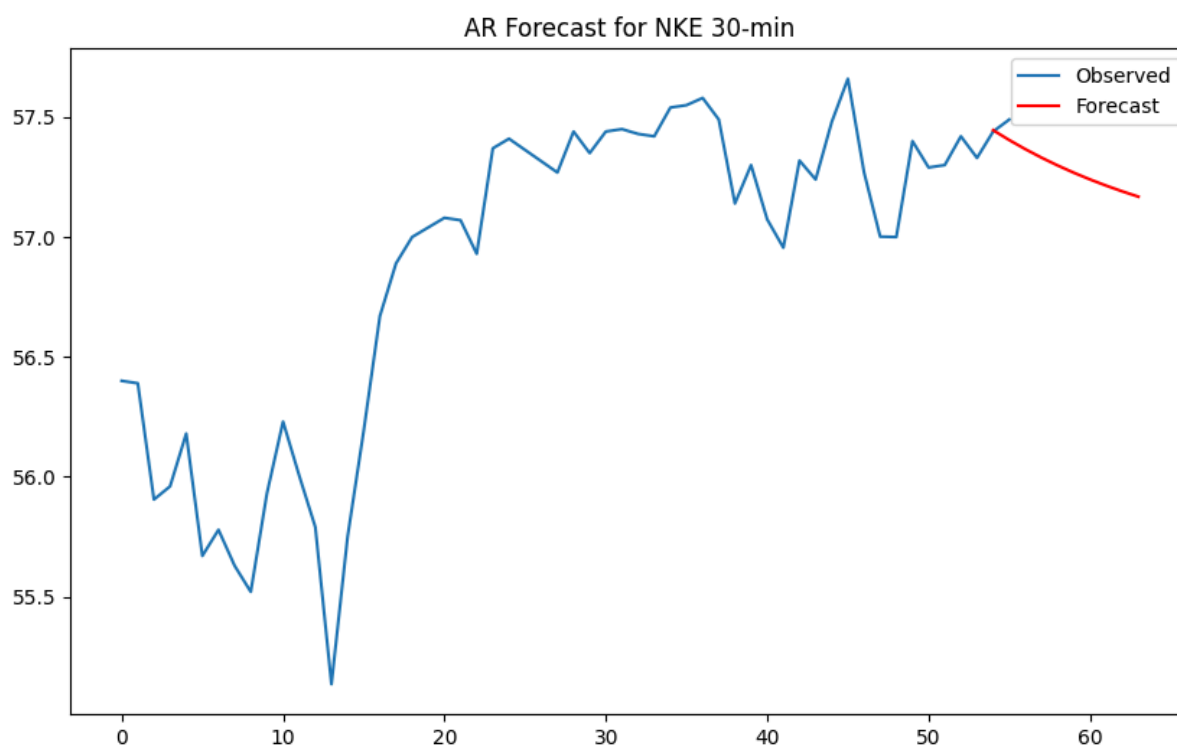
Prob(H) (two-sided): 0.05 Kurtosis: 3.65

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARMA Model:

SARIMAX Results

```
=====
Dep. Variable:      Close  No. Observations:      54
Model:              ARIMA(1, 0, 1)  Log Likelihood      -2.436
Date:              Fri, 02 May 2025  AIC              12.872
Time:              23:15:15  BIC              20.828
Sample:            0  HQIC              15.940
                    - 54
```

Covariance Type: opg

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const    56.8923    0.451  126.235   0.000   56.009   57.776
ar.L1     0.9175    0.053   17.174   0.000    0.813    1.022
ma.L1     0.0577    0.137    0.420   0.675   -0.211    0.327
sigma2    0.0618    0.011    5.556   0.000    0.040    0.084
=====
```

```
=====
Ljung-Box (L1) (Q):      0.00  Jarque-Bera (JB):      3.95
Prob(Q):                 0.98  Prob(JB):              0.14
```

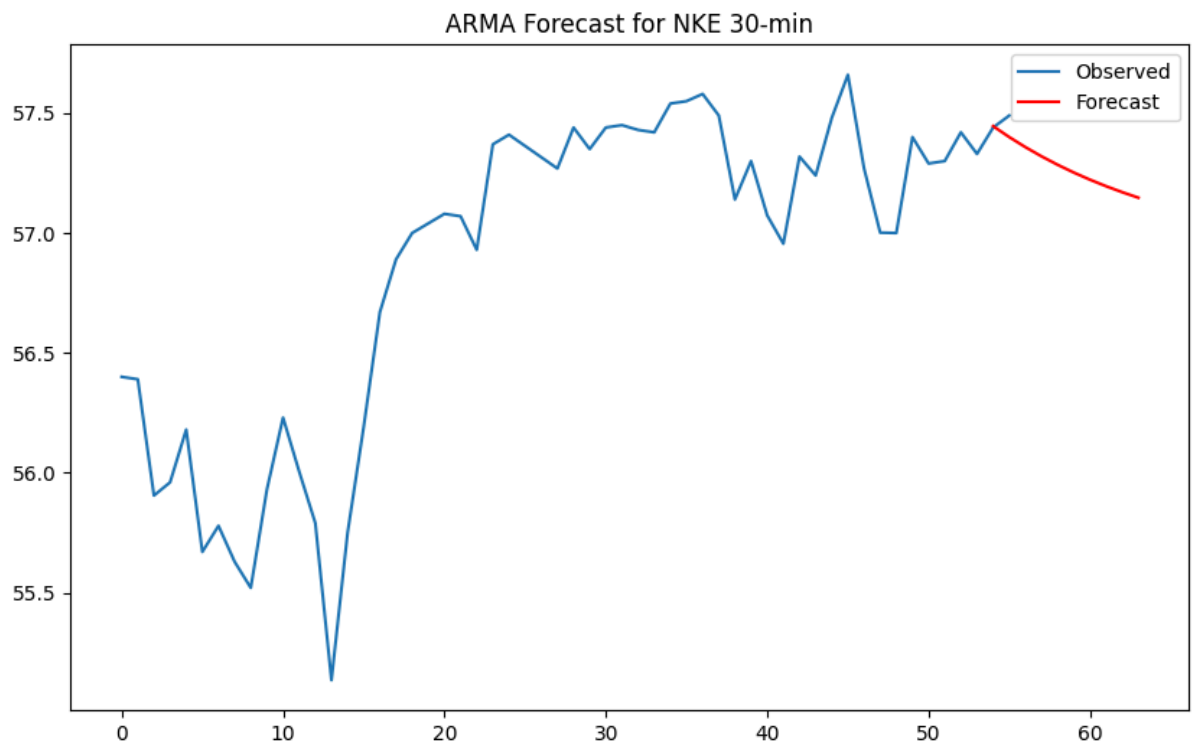
Heteroskedasticity (H):	0.39	Skew:	-0.57
Prob(H) (two-sided):	0.05	Kurtosis:	3.68

=====

=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



ARIMA Model:

SARIMAX Results

=====

Dep. Variable:	Close	No. Observations:	54
Model:	ARIMA(1, 1, 1)	Log Likelihood	-2.352
Date:	Fri, 02 May 2025	AIC	10.704
Time:	23:15:15	BIC	16.615
Sample:	0	HQIC	12.977
	- 54		

Covariance Type: opg

=====

coef	std err	z	P> z	[0.025	0.975]
------	---------	---	------	--------	--------

ar.L1	0.6816	0.711	0.959	0.337	-0.711	2.074
ma.L1	-0.7519	0.689	-1.092	0.275	-2.101	0.598
sigma2	0.0640	0.012	5.343	0.000	0.040	0.087

=====

=====

Ljung-Box (L1) (Q): 0.19 Jarque-Bera (JB): 0.81

Prob(Q): 0.67 Prob(JB): 0.67

Heteroskedasticity (H): 0.37 Skew: -0.26

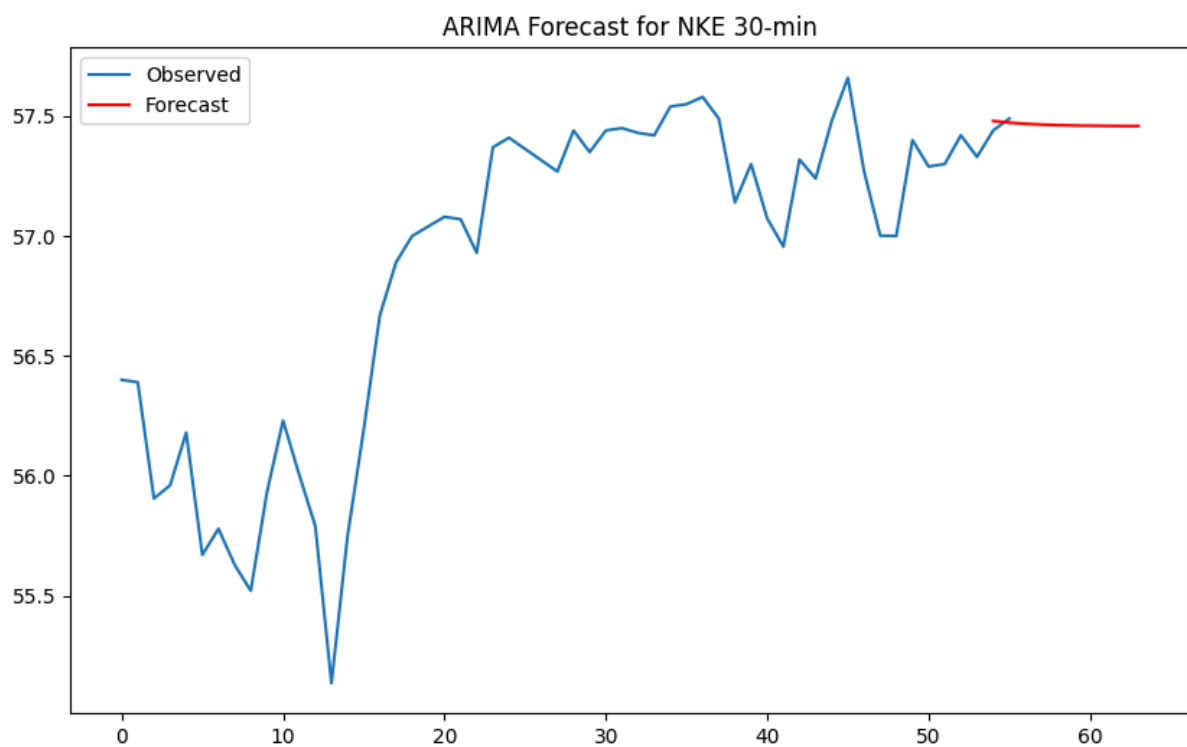
Prob(H) (two-sided): 0.04 Kurtosis: 3.31

=====

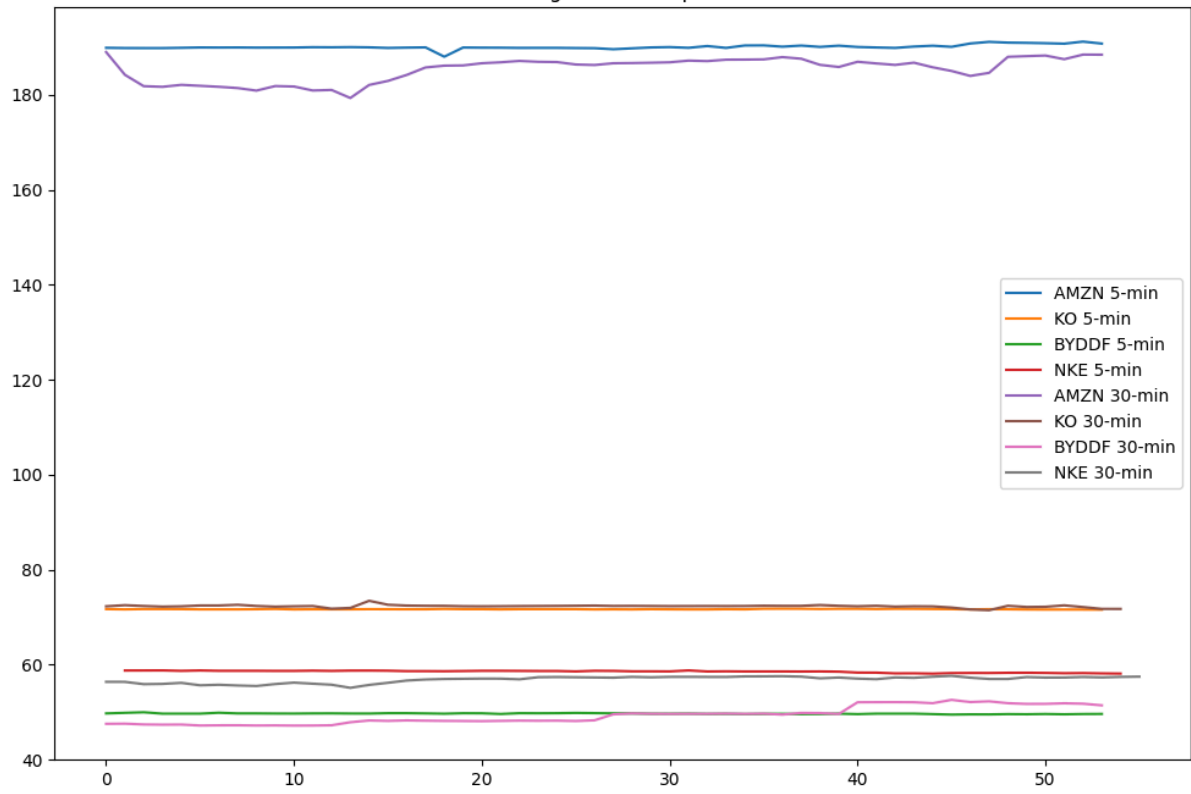
=====

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Closing Prices Comparison



Estrategia de portafolio:

KO (Coca-Cola) – Posición Larga

Razonamiento: La más estable y predecible según los tests de estacionariedad y buen ajuste del modelo.

Asignación: \$25,000 para aprovechar la estabilidad en el corto plazo.

✓ ORDER PLACED SUCCESSFULLY

Your Market Buy for 340 shares of KO (Coca-Cola Co) will be processed at the next market open.

[Trade Again](#) [View Order](#)

KO				10	68.56	71.65	0.36	\$716.50	30.90		TRADE
Coca-Cola Co									(4.51%)		

NKE (Nike) – Posición Larga

Razonamiento: Fuerte tendencia con buena respuesta en el modelo ARIMA.

Asignación: \$25,000 para capturar el impulso mientras siga la tendencia.

NKE				10	54.28	58.59	1.83	\$585.90	43.10		TRADE
Nike, Inc....									(7.94%)		

5/2/2025

Market - Buy

NKE

400

MKT

-

EQUITIES

USD

83087f0e-5721-4e21-8a12-a90c8b2d7cdc

CANCEL

EDIT

ADD/VIEW NOTES

AMZN (Amazon) – Vender

Razonamiento: Alta volatilidad y comportamiento inestable; se busca aprovechar movimientos a la baja.

Asignación: Vender las 10 acciones con las que cuento.

5/2/2025

Market - Sell

AMZN

-10

MKT

-

EQUITIES

USD

54ea16a4-962c-45fc-8439-fd0a1c971c84

CANCEL

EDIT

ADD/VIEW NOTES

BYDDF (BYD Co.) – Vender

Razonamiento: Mucha heterocedasticidad y poca confianza para largos, pero se puede tomar una posición pequeña.

Asignación: Vender las 10 acciones con las que cuento.

✓ ORDER PLACED SUCCESSFULLY

Your Market Sell for -10 shares of BYDDF (BYD Co - Ordinary Shares - Class H) will be processed at the next market open.

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Control de Riesgos – Estrategia Actual

Para las posiciones largas en KO y NKE estoy metiendo \$50,000 en total, repartidos en \$25,000 para cada una. Mi idea es poner un stop loss del 5%, o sea, si alguna llega a caer unos \$1,250, la vendo para evitar perder más. También tengo pensado asegurar ganancias si suben alrededor de un 10%. Voy a estar revisando el mercado mínimo cada hora para ver si la tendencia sigue firme y, si veo que el modelo empieza a fallar o hay señales raras, salgo antes aunque no haya tocado el stop.

En el caso de AMZN y BYDDF, la estrategia es simple: vender ya las 10 acciones que tengo de cada una porque no me da confianza seguir ahí. La idea es liberar ese capital y reducir riesgos, aunque voy a seguir viendo cómo se mueven por si más adelante hay chance de entrar otra vez si aparecen oportunidades claras, quería hacerles short pero no me dejo la aplicación debido a que ya está cerrado el mercado.

Todo lo que recupere de esas ventas lo voy a dejar líquido o en algo súper seguro mientras veo qué más conviene hacer. La idea es no sobreexponer y mantener la mitad del portafolio libre para poder moverme rápido si el mercado cambia o si salen nuevas oportunidades.

Por último, si llega a pasar algo fuerte a nivel macroeconómico o alguna noticia importante que afecte al mercado, estoy listo para cerrar todas las posiciones de inmediato aunque no se haya activado el stop loss, siempre priorizando proteger el capital.

Evaluación de Resultados y Reflexiones Finales

Lo que espero con esta estrategia es que KO mantenga su estabilidad y me dé un rendimiento constante en el corto plazo, aprovechando esos movimientos predecibles que ya vi en el análisis. En el caso de NKE, la idea es que la tendencia siga fuerte para poder capturar ganancias mientras la acción se mantenga al alza, aunque sé que puede ser un poco más volátil.

Con las ventas de AMZN y BYDDF, lo que busco es reducir riesgos desde el inicio y quedarme solo con las posiciones que considero más seguras o con mejor potencial. Espero que esto me ayude a proteger el capital y tener mayor flexibilidad si se presentan nuevas oportunidades más adelante.

En general, espero que esta estrategia me sirva para mejorar mi control del riesgo y para darme más confianza en tomar decisiones basadas en datos, no solo en intuición. También quiero comprobar si lo que vi en los modelos de series de tiempo se refleja

realmente en los resultados de la plataforma, y seguir afinando mis análisis para ser más preciso en futuras inversiones.

Interpretación de las Pruebas de Hipótesis, Conclusiones y Recomendaciones

Antes de ver los resultados futuros, las pruebas de hipótesis me dejaron claro que KO es la acción más estable y predecible, ya que pasó bien las pruebas ADF y KPSS. NKE, aunque no es estacionaria, mostró una tendencia fuerte y una autocorrelación bastante buena, lo cual me da confianza para apostar por la continuidad de su impulso. En cambio, AMZN y BYDDF no pasaron las pruebas de estacionariedad y mostraron alta volatilidad, por lo que decidí no mantener esas posiciones y reducir riesgos desde el inicio.

En conclusión, mi estrategia está bien respaldada por los datos y espero que las posiciones en KO y NKE me permitan aprovechar movimientos más controlados y con mayor probabilidad de éxito. Vender AMZN y BYDDF fue una decisión más enfocada en proteger capital y estar listo para futuras oportunidades.

Como recomendación, creo que es clave seguir aplicando estos análisis técnicos antes de cada operación y no confiar solo en la intuición. También considero importante mantener un control estricto de los stop loss y seguir monitoreando las posiciones para hacer ajustes rápidos si el comportamiento cambia.